

Master's Thesis

Design, Implementation and Evaluation of a Digital Shadow for a Simple Bioreactor for State Estimation and Fault Identification

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This work presents a general road-map for creating a *digital shadow* (DS) of a simple bioreactor and demonstrates, through a working example, a method for performing fault detection and root cause analysis (RCA) on top of that shadow. The digital shadow implemented by [YPH21] is first reproduced and then extended with three additional environmental states — temperature, pH and dissolved oxygen — and a fault-detection and root-cause-analysis layer is built on top of this model. The estimation core is an Extended Kalman Filter driven by a single ethanol off-gas measurement. The main findings are that this single sensor is sufficient to reconstruct the internal states within the limits set by observability — and that a cumulative-sum (CUSUM) test on the ethanol innovation detects growth-affecting faults, whose cause is then isolated by a bank of hypothesis filters (multiple-model adaptive estimation, MMAE).

Chapter 1 reviews the main concepts — the digital shadow, the digital twin and root cause analysis — and the problem this work solves (Section 1.1); it then surveys related work on digital twins and shadows in particular digital twins in healthcare and in RCA (Section 1.2), before explaining the motivation (Section 1.3). Chapter 2 presents the working example: a batch bioreactor whose estimation core is the ethanol-driven Extended Kalman Filter, its implementation in Python, and its extension into a CUSUM-based fault detector with MMAE root-cause analysis. Chapter 3 reports the results, Chapter 4 evaluates the final framework, and Chapter 5 discusses conclusions, limitations and recommendations for future work.

1 Introduction

Introduction and related concepts

A *Digital Twin* (DT) is a dynamic virtual representation of a physical system that is continuously updated with real-time data from sensors, IoT devices, and other monitoring sources, enabling analysis, simulation, and prediction of a real-world object or process [Sad+24]. The concept usually involves three parts: a physical entity in the real world, a digital representation in a virtual world, and information exchange between the two [Sag23]. The degree of that information exchange is what distinguishes the related concepts. Following the widely adopted categorization of Kritzinger et al. [Kri+18], a *Digital Model* has no automated data exchange, a *Digital Shadow* (DS) has an automated data flow *from* the physical object *to* the digital object, and a full *Digital Twin* closes the loop with automated data exchange in both directions. Because the objective of this work is anomaly detection and root cause analysis, monitoring and diagnosis, a digital shadow is the appropriate and sufficient architecture: the physical bioreactor drives the virtual replica in real time, while corrective decisions remain with the operator.

Maintenance of complex processes remains a challenge across industries, and firms are increasingly moving toward predictive-maintenance strategies to reduce unplanned downtime, failures, and accidents [Fan+22]. In this setting, digital twins and shadows are attractive because, in practice, they simulate the ideal (failure-free) behavior of an asset and continuously compare it against real-world data, allowing the analytical layer to flag deviations caused by wear, parameter drift, or environmental factors [HB26]. *Root cause analysis* (RCA) is the systematic tracing of such observed anomalies back to their underlying cause [Wil01]; in a DS context it forms the bridge between data-driven diagnostics and operational decision-making, because the idealized model provides the reference against which residuals become interpretable [HSB24].

Fault diagnosis built on a digital twin is a rapidly maturing field. Recent reviews show that DT-based intelligent fault diagnosis has become a distinct research direction, precisely because the twin supplies a physics-consistent baseline and abundant simulated fault data that purely data-driven detectors otherwise lack [Hu+25].

Applying these ideas to a bioreactor is, however, distinctly harder than to a mechanical asset. The process is a living, non-linear, and only partially observable system: the biologically most relevant quantities — biomass, substrate, and product concentrations — are rarely measured on-line, robust probes exist mainly for physical variables such as temperature, pH, and dissolved oxygen, and normal batch-to-batch variability is large enough to mask incipient faults. Luo et al. [Luo+24] document exactly these difficulties for fermentation processes and argue for a “golden-batch”-

driven reference against which anomalies in a bioreactor are detected, which is conceptually the same reference role played by the digital shadow in this thesis. Existing RCA practice in bioprocessing has typically relied on multivariate statistical workflows — raw-data and feature-based analyses using principal component analysis and partial least squares — which can identify the variables associated with a deviation but are limited by their dependence on historical data, their weak grasp of the underlying mechanism, and their difficulty attributing a residual to a specific physical cause [Bor+19]. Data-driven twins for cell-culture forecasting share the same limitation: they extrapolate poorly outside their training distribution and offer little mechanistic explanation of *why* a batch deviates [Par+23]. Model-based fault diagnosis reviews reach a similar conclusion, noting that data-only approaches struggle with interpretability and with faults unseen during training [Hu+25]. These gaps motivate a mechanistic, estimator-centered digital shadow whose residuals are physically meaningful, which is the approach developed here.

In summary, this work develops a digital shadow of a simple bioreactor that enables the continuous estimation of the ethanol, glucose and biomass concentrations, as well as of temperature, pH and dissolved oxygen. In its second part, this shadow is extended to detect anomalies and to trace them back to their root cause. All the result reported in this work are reproducible using the implementation in the Git Repository.

Background

Digital shadows and twins in healthcare.

Although this thesis targets a bioreactor, much of the maturity of the DT/DS concept comes from healthcare, and that body of work motivates the design choices made here. In healthcare, digital twins have been explored for predictive monitoring, personalized treatment, surgical planning, optimization of care workflows, and drug discovery [Hua+22], and a consortium-level account describes the concepts and applications of digital twins across medicine [Zha+24]. The most significant contributions have been in precision medicine, where a detailed patient model reproduces biological processes and predicts their evolution under treatment [Sad+24; Kar+20]. When integrated with AI and machine learning, DT technology is argued to hold substantial potential in oncology [GM25], and health digital twins are increasingly framed as tools for precision medicine [VRK22]. A frequently cited secondary benefit is the reduction of animal experimentation, since a validated model can translate *in vitro* measurements into expected *in vivo* behavior [MUB23]. Two lessons carry over to the bioreactor problem: first, that the highest-value twins are those that estimate quantities which cannot be measured directly; and second, that most successful implementations model a single, well-scoped unit rather than an entire system [Ria+22], which is the scoping strategy adopted in this thesis.

Digital shadows for root cause analysis.

A digital shadow supports RCA by supplying the idealized, failure-free reference trajectory against which live data are continuously compared, so that deviations become detectable and, in turn, attributable [HB26]. RCA itself is a mature discipline with a standard toolkit — among others the five-whys technique, fault-tree analysis, and cause-and-effect diagrams [Wil01]. What a model-based shadow adds is a physically meaningful residual: instead of correlating raw variables, the estimator produces an innovation sequence whose statistical behavior can be tied to specific physical faults [VCM12; DSG14], and competing physical fault hypotheses can be scored against that residual [HM00]. This is the conceptual foundation of the RCA layer built on top of the bioreactor shadow in Section 4.4.

Motivation

The motivation for this work is threefold. First, it addresses a real observability gap: in a yeast bioreactor the states that matter most for product quality — biomass, substrate, and product concentrations — are expensive or hard to measure continuously on-line, whereas the estimator-based shadow reconstructs them from a single cheap off-gas signal [YPH21], turning one affordable sensor into a full internal state estimate. Second, it targets predictive maintenance and diagnosis: by continuously comparing the shadow’s failure-free prediction with live data, incipient faults — a failing heater, a stuck pH pump, an aeration fault — can be detected and, through the RCA layer, traced to their cause [HB26; Luo+24], reducing the risk of losing an entire multi-day batch. Third, it contributes methodological value: part of the bioprocess DT literature remains conceptual [Hua+22; Mua+24], and a concrete, reproducible, mechanistically interpretable shadow with a built-in diagnosis mechanism helps close the gap between architecture papers and working implementations.

The intended beneficiaries are therefore process engineers and operators of small- to mid-scale cultivation equipment who need continuous insight into an under-instrumented process without the cost of a full analytical sensor suite; biomanufacturers pursuing Quality-by-Design, for whom a mechanistic shadow provides both monitoring and an auditable diagnostic trail [Mua+24]; and researchers who can reuse the road-map and the open estimator as a template for other cultivation systems. Because the approach relies on a validated process model rather than a large historical dataset, it is also applicable in early-stage process development, where data are scarce and each experimental run is costly.

In the hands of domain experts, and with access to laboratory experiments, the proposed road-map — together with the implemented digital shadow and the fault-detection and root-cause-analysis layer built on top of it — has the potential to be applied in real experiments and in industrial settings.

2 Methodology

A Digital Shadow for Baker’s Yeast Cultivation

The design of a digital shadow for a bioreactor depends on several factors. The most obvious are the type of bioreactor and its operating envelope, and the organism being cultivated together with its growth mechanism. For the RCA component one must additionally enumerate what can go wrong in the process: which fault scenarios are plausible, and how each perturbs the measurable environment of the reactor. This section first fixes the necessary bioprocess background and then sets out the shadow architecture; the concrete yeast realization follows in the subsequent sections.

What is a Bioreactor

A bioreactor is a vessel in which a biological reaction is carried out under controlled conditions, providing a defined physical and chemical environment — temperature, pH, dissolved oxygen, and mixing — for the growth of cells or the action of enzymes, so that a desired product is formed [Dor13]. For modelling purposes the workhorse abstraction is the ideally mixed stirred-tank reactor, in which the contents are assumed spatially homogeneous so that the process can be described by a set of ordinary differential equations coupling a biokinetic sub-model to physico-chemical balances for pH and oxygen transfer and to a reactor sub-model [Mos+21]. In this work the reference hardware is a laboratory-scale stirred tank of the Infors Minifors class, whose heated jacket, aeration and agitation system, and pH-control pumps define both the physical balances that the shadow must reproduce and the actuators whose failure constitutes the fault scenarios considered later. An example is shown in Figure 2.1.

What is Cell Cultivation

Cell cultivation is the controlled propagation of cells in such a vessel by supplying substrate and maintaining a favourable environment. This thesis focuses on the batch cultivation of baker’s yeast, *Saccharomyces cerevisiae*, whose metabolism is characteristically *diauxic*: under aerobic batch conditions the cells first consume glucose — producing ethanol by overflow metabolism even in the presence of oxygen — and, once glucose is depleted, switch to re-consuming the ethanol they previously excreted [PLE98]. This overflow-and- switch behaviour is the reason ethanol is such an informative measurement: its rise reports the glucose-consumption phase and its subsequent decline reports the ethanol-consumption phase, so a single ethanol channel carries information about the entire batch trajectory. The kinetics are captured with Monod-type growth expressions that include a diauxic suppression term, and

the maximum specific growth rates are treated as slowly varying quantities that differ between strains and batches — which is precisely why an *adaptive* estimator, rather than a fixed forward simulation, is required to track a real cultivation.

A Digital Shadow for a Bioreactor

Several approaches exist for simulating a physical system through a one-directional information flow, some of which were discussed in Section 1.2. What is required here is a digital shadow of the bioreactor that operates in real time: it must accept updates from the real world through the available sensors and, at the same time, estimate the state of every relevant internal variable from knowledge of how the real system behaves. Only when both capabilities are present can the shadow first recognize that the real system has become abnormal and then, through an RCA mechanism, propose a probable cause for that abnormal behavior [HB26; Luo+24].

The computational core of such a shadow is therefore a state estimator that fuses a mechanistic process model with sparse, noisy measurements. The Kalman Filter is the classical choice for this task and is, for a linear system with additive white Gaussian noise, the optimal minimum-variance estimator [Kal60; Sim06]. A bioreactor is non-linear, e.g. in presence of double glucose the growth rate will not double, so the strictly optimal linear result does not apply directly; one of the well-known engineering solution is the Extended Kalman Filter, which linearises the process model around the current estimate at each step, or the Unscented Kalman Filter, which avoids that linearization by propagating sample points [WM00]. Both extensions have a strong track record in the supervision of cultivation processes [YPH21], and an ethanol-driven Extended Kalman Filter has been shown to reconstruct the states of a baker’s-yeast batch from a single gas-sensor signal [YPH21]. For these reasons, and because the same filtering framework simultaneously provides the innovation sequence needed for fault detection and root cause analysis [VCM12; DSG14; HM00], the digital shadow developed in this work is built on an Extended Kalman Filter: it delivers continuous state estimates under the operating assumptions of the process while doubling as the residual generator for the RCA layer.

Standard Kalman Filter

The Kalman Filter (KF) is a set of mathematical equations that estimates the state of a dynamic system at arbitrary time points [Kal60]. It is particularly useful when measurements are not continuously available, leaving gaps between data points that the filter fills by propagating a mathematical model forward in time. It can be shown that under the assumption of Gaussian noise and a linear system, the KF provides the mathematically optimal state estimate in the minimum mean square error sense.

The KF requires a model describing how the state vector $\mathbf{x}$ evolves over time:

$$\mathbf{x}_{new} = A \cdot \mathbf{x}_{old} \quad (2.1)$$

where A is the state transition matrix derived from the known physics or biology of the system. For example, if $\mathbf{x}$ represents the temperature of a closed system, A

can be set from the heat transfer law.

Since no model is perfect, the KF maintains an uncertainty matrix P that tracks how confident the filter is in its current estimates. During prediction, P grows according to:

$$P_{new} = A \cdot P_{old} \cdot A^T + Q \quad (2.2)$$

where Q is the process noise covariance — a fixed matrix expressing how much the model is expected to deviate from reality per time step. It is important to note that Q appears in the covariance equation only, not in the state equation: it does not directly shift the state estimate but increases the uncertainty P , allowing the filter to remain open to correction by future measurements.

Whenever a sensor measurement z arrives, the KF performs a correction step. The measurement is modelled as:

$$z = H \cdot \mathbf{x}_{true} + v, \quad v \sim \mathcal{N}(0, R) \quad (2.3)$$

where H is the observation matrix that maps the state vector to the measured quantity, and v is measurement noise with covariance R . If the measured quantity corresponds directly to a state variable, H contains a one in the corresponding position and zeros elsewhere. If a unit conversion is required — for example measuring temperature in Celsius while the model uses Kelvin — H encodes the conversion factor.

The corrected state estimate is:

$$\mathbf{x}_{new} = \mathbf{x}_{old} + K \cdot (z - H \cdot \mathbf{x}_{old}) \quad (2.4)$$

where $(z - H \cdot \mathbf{x}_{old})$ is the innovation — the difference between the measured value and the model prediction — and K is the Kalman gain:

$$K = P \cdot H^T \cdot (H \cdot P \cdot H^T + R)^{-1} \quad (2.5)$$

The Kalman gain balances the relative trust between the model prediction and the measurement. When the measurement noise R is large, K becomes small and the filter relies more heavily on its model prediction. When the uncertainty P is large — meaning the model has been propagating without measurements for a long time — K becomes large and the filter trusts the incoming measurement more. After each correction step, P decreases, reflecting the information gained from the measurement.

Why the Standard KF Cannot Be Applied Directly

The Kalman Filter (KF) provides an efficient recursive solution of the least-squares estimation problem, but it is optimal only under two assumptions: that the system dynamics are *linear* and that the process and measurement uncertainties are zero-mean Gaussian random variables [YPH21; Kal60; Sim06]. Under these assumptions the process is written as the linear recursion $\mathbf{x}[k] = A \mathbf{x}[k-1] + B \mathbf{u}[k-1] + \mathbf{w}[k-1]$, in which the fixed matrix A relates the previous state to the current one and B maps

the control input onto the states. A biological cultivation, however, is a non-linear and complex system, and Yousefi-Darani, Paquet-Durand, and Hitzmann [YPH21] note explicitly that for such systems the process model cannot be expressed as a multiplication of fixed matrices and vectors, so that a linearization step becomes unavoidable.

This work inherits exactly that difficulty. The bioreactor process model of Section 2.1.4 violates the linearity requirement in several places. The Monod growth expressions multiply an estimated parameter by a non-linear function of a state; for example

$$\mu_G = \mu_{max,G} \cdot \frac{G}{K_G + G} \quad (2.6)$$

couples $\mu_{max,G}$ with the saturating term $G/(K_G + G)$. Likewise, the biomass balance $dX/dt = (\mu_G + \mu_E) \cdot X$ multiplies the (state-dependent) growth rates by the biomass concentration itself. These bilinear and saturating interactions cannot be represented by any single constant matrix A acting on the state vector, so the standard KF in its linear form is inapplicable and a non-linear extension is required.

Extended Kalman Filter

For non-linear cultivation models the field has converged on the non-linear extensions of the Kalman Filter, chiefly the Extended Kalman Filter (EKF) and the Unscented Kalman Filter (UKF), together with the ensemble (EnKF) and cubature (CKF) variants for special cases [YPH21]. The variants differ only in *how* they propagate the estimation error through the non-linearity. The EKF linearises the model about the current mean and covariance by computing the Jacobian matrix

$$F = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\mathbf{x}_{current}}, \quad (2.7)$$

where $f(\mathbf{x})$ is the non-linear process model (Equation 2.9). Geometrically, F is the local slope of the non-linear dynamics at the current operating point — analogous to the tangent of a curve — and, unlike the fixed matrix A of the linear KF, it is recomputed at every time step as the state evolves. In the formulation adopted here the state $\mathbf{x}$ is always propagated with the *full* non-linear model, integrated numerically with a Runge–Kutta solver, and only the covariance propagation uses the linearised Jacobian:

$$\frac{dP}{dt} = F \cdot P + P \cdot F^T + Q. \quad (2.8)$$

This is precisely the continuous–discrete EKF of Yousefi-Darani, Paquet-Durand, and Hitzmann [YPH21]: the biology is integrated continuously between measurements while the ethanol sensor delivers a discrete correction every five minutes, matching the physical structure of the process.

Comparison of the filter variants and rationale for the EKF.

The choice of the EKF over the other non-linear variants follows the evidence assembled by Yousefi-Darani, Paquet-Durand, and Hitzmann [YPH21]. In their survey of Kalman-filter applications to bioprocesses (41 studies over 1991–2020), more than 60% of the implementations (28 of the reported works) used the EKF, precisely because microbial cultivation is a complex non-linear process for which the EKF is the established, well-understood estimator. The decisive practical argument is cost and the simplicity of implementation: the EKF is attractive mainly because of its *simple structure and low computational effort*, requiring only the Jacobian and a single propagation of the mean rather than the multiple sample-point evaluations of the UKF, EnKF or CKF. The EnKF and CKF only become worthwhile when the model is of extremely high order, the initial state is highly uncertain, and a large number of measurements are available [YPH21]; for mildly non-linear systems the EKF is a common and popular choice [Zal25]. Since none of these conditions holds for the low-order, ethanol-driven yeast model the EKF is used here.

The genuine weakness of the EKF is equally documented and worth stating honestly. Because it propagates only the mean through the non-linearity, the EKF is not optimal for *highly* non-linear systems [WM00]; it is reliable only for models that are approximately linear on the time scale of the update interval, it requires a Jacobian at every step, and its linear approximation can accumulate error and eventually diverge. A concrete failure mode reported in the chapter is that, in fed-batch cultivations where the substrate is consumed almost as fast as it is fed, the EKF can assign a non-zero probability to negative substrate concentrations, which is physically impossible, and the UKF has been shown to be more accurate in that regime [YPH21]. Two considerations make the EKF the right choice for the present work nonetheless. First, the reference process is a *batch* cultivation of *S. cerevisiae*, not a fed-batch one, so the negative-substrate pathology that motivates the UKF is far less pressing, and the model is well approximated by linearisation on the five-minute update scale. Second, the negative-concentration artefact is suppressed directly in the implementation (Section 2.2.1) by clamping all concentrations to non-negative values at every sub-step, which removes the specific EKF weakness the chapter warns about while retaining its low computational cost.

For these reasons, and because the same filtering framework simultaneously yields the innovation (residual) sequence that the later fault-detection and root-cause-analysis layers rely on, the digital shadow developed here is built on the continuous–discrete EKF. It reproduces the estimator of Yousefi-Darani, Paquet-Durand, and Hitzmann [YPH21]. Because that work already demonstrated a functioning EKF-based digital shadow of the baker’s-yeast cultivation process, and because its scope — online state estimation of an under-instrumented yeast batch — matches the scope of this thesis, it is adopted as the primary reference for the implementation. The present work then *extends* that shadow by adding further environmental states, by evaluating hypotheses such as how much an additional sensor improves estimation and diagnosis accuracy, and by adding a fault-detection layer together with a working root-cause-analysis procedure on top of the original estimator.

Process model

The biological dynamics of the batch cultivation are described by the process model of Yousefi-Darani, Paquet-Durand, and Hitzmann [YPH21], which consists of a system of five ordinary differential equations (Equation 19 in the reference paper):

$$\frac{d}{dt} \begin{pmatrix} X \\ G \\ E \\ \mu_{max,G} \\ \mu_{max,E} \end{pmatrix} = \begin{pmatrix} (\mu_G + \mu_E) \cdot X \\ -\frac{\mu_G}{Y_{gx}} \cdot X \\ \left(\frac{\mu_G \cdot Y_{ge}}{Y_{gx}} - \frac{\mu_E}{Y_{ex}} \right) \cdot X \\ 0 \\ 0 \end{pmatrix} \quad (2.9)$$

The state vector $\mathbf{x} \in \mathbb{R}^5$ contains five state variables:

$$\mathbf{x} = \begin{pmatrix} X & G & E & \mu_{max,G} & \mu_{max,E} \end{pmatrix}^T \quad (2.10)$$

where X is the biomass concentration [g/L], G is the glucose concentration [g/L], E is the ethanol concentration [g/L], $\mu_{max,G}$ is the maximum specific growth rate on glucose [h⁻¹], and $\mu_{max,E}$ is the maximum specific growth rate on ethanol [h⁻¹].

It is evident from Equation 2.9 that the state derivatives depend on the actual growth rates μ_G and μ_E , which are not state variables themselves but are computed at each time step from the current state using Monod kinetics [Mon49]:

$$\mu_G = \mu_{max,G} \cdot \frac{G}{K_G + G} \quad (2.11)$$

$$\mu_E = \mu_{max,E} \cdot \frac{E}{K_E + E} \cdot \left(1 - \frac{\mu_G}{\mu_{max,G}} \right) \quad (2.12)$$

Monod kinetics is a mathematical model describing how the specific growth rate of a microorganism depends on the concentration of a limiting substrate [Mon49]. At high substrate concentrations the growth rate approaches its maximum value μ_{max} ; as the substrate becomes scarce the rate decreases and approaches zero. The half-saturation constant K defines the substrate concentration at which the growth rate is exactly half of its maximum. For both glucose and ethanol, $K_G = K_E = 0.1$ g/L [YPH21].

The second Monod term in Equation 2.12 contains the suppression factor $(1 - \mu_G/\mu_{max,G})$, which models the diauxic growth behavior of *S. cerevisiae*. When glucose is abundant and $\mu_G \approx \mu_{max,G}$, this factor approaches zero and ethanol consumption is completely suppressed. Only when glucose is exhausted and $\mu_G \rightarrow 0$ does the suppression lift and the cells switch to ethanol as their carbon source.

The last two rows of Equation 2.9 state $d\mu_{max,G}/dt = 0$ and $d\mu_{max,E}/dt = 0$. This does not mean these parameters are constant — it means the process model provides no dynamics for them. They are treated as slowly varying unknown parameters whose values are updated exclusively through the Kalman correction step when

measurements arrive, as described in Section 2.1.4.

The yield coefficients Y_{gx} , Y_{ge} , and Y_{ex} describe the stoichiometry of the cultivation. The values used in the implementation follow the MATLAB code provided in the appendix of Yousefi-Darani, Paquet-Durand, and Hitzmann [YPH21]: $Y_{gx} = 0.15$ g/g (biomass yield from glucose), $Y_{ge} = 0.34$ g/g (ethanol yield from glucose), and $Y_{ex} = 0.43$ g/g (biomass yield from ethanol). It should be noted that these values differ from those reported in Table 2 of the paper ($Y_{X/G} = 0.17$, $Y_{E/G} = 0.46$, $Y_{X/E} = 0.60$); the code values were adopted as they produced results more consistent with the published figures.

Observation Model

In this implementation, the only available online measurement is the ethanol concentration E , which corresponds to the third element of the state vector. The observation matrix H is therefore:

$$H = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \end{pmatrix} \quad (2.13)$$

The measurement equation becomes:

$$z_k = E(t_k) + v_k, \quad v_k \sim \mathcal{N}(0, R) \quad (2.14)$$

where $R = 0.1$ (g/L)² is the measurement noise covariance, set to reflect the uncertainty of the gas sensor pipeline.

Initialization

All initial values were set based on prior knowledge from previous experiments with baker's yeast under similar conditions [YPH21]. The initial state vector was:

$$\mathbf{x}_0 = \begin{pmatrix} 2.5 & 6.0 & 0.2 & 0.15 & 0.08 \end{pmatrix}^T \text{ [g/L, g/L, g/L, h}^{-1}\text{, h}^{-1}] \quad (2.15)$$

The initial glucose estimate follows the value in the MATLAB code accompanying [YPH21] ($G_0 = 6$ g/L), which differs from the ≈ 5 g/L shown in that paper's figures (Fig. 4, [YPH21]); the true simulated batch in this work starts at 5 g/L.

The initial covariance matrix P_0 was set as a diagonal matrix reflecting the uncertainty in each initial estimate:

$$P_0 = \text{diag}(0.1, 0.02, 0.02, 0.2, 0.02) \quad (2.16)$$

The largest value (0.2) is assigned to $\mu_{max,G}$, as the maximum specific growth rate is the most difficult parameter to know a priori from initial conditions alone. The process noise covariance was set to:

$$Q = \text{diag}(0.001, 0.001, 0.001, 0.001, 0.001) \quad (2.17)$$

reflecting equal and small model uncertainty across all state variables.

Covariance Update — Joseph Form

The covariance matrix after each correction step was updated using the numerically stable Joseph form:

$$P_{new} = (I - KH) \cdot P \cdot (I - KH)^T + K \cdot R \cdot K^T \quad (2.18)$$

This formulation is mathematically equivalent to the standard update $P_{new} = (I - KH) \cdot P$ but remains positive definite even when Q and R are not perfectly calibrated, improving numerical robustness.

The Physical System

The physical system investigated in this work, as illustrated in Figure 2.1, a 2.5 L stirred tank (Minifors, Infors HT, Bottmingen, Switzerland) bioreactor for the batch cultivation of baker's yeast (*Saccharomyces cerevisiae*).

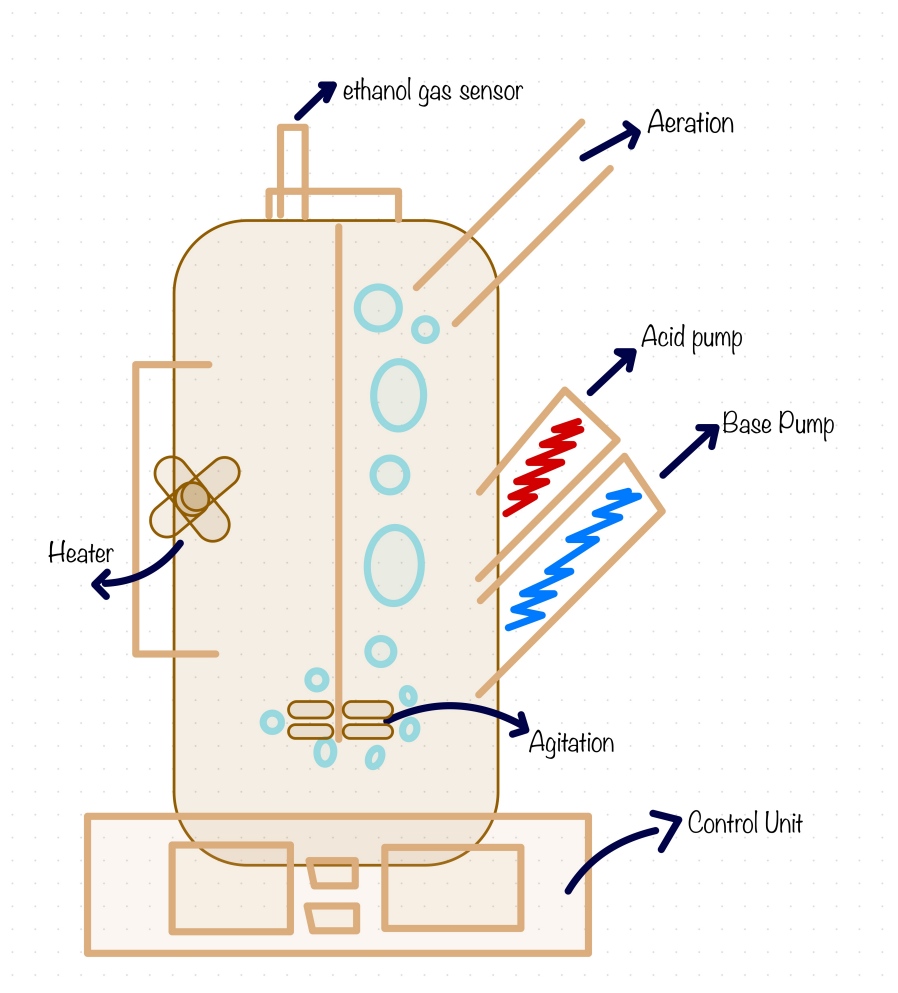


Figure 2.1: Overview of the experimental setup. inspired by Fig.2 [YPH21]

The bioreactor is inoculated with yeast cells and an initial glucose concentration

of 5.0 g/L. Throughout the experiment, temperature is maintained at 30°C, pH is controlled at 5.0, aeration is set to 3.5 L/min, and agitation is maintained at 500 rpm. The process is conducted aerobically — oxygen is continuously supplied by the aeration system. The total batch duration is approximately seven hours.

The cultivation proceeds in two distinct phases. During the first phase (approximately 0 h to 2 h), the yeast cells consume glucose as their primary carbon source. As a result of the Crab-tree effect — a metabolic characteristic of *S. cerevisiae* — ethanol and CO₂ are produced simultaneously even under aerobic conditions. When the glucose is exhausted at approximately $t = 2$ h, the second phase begins: the cells switch to ethanol as their sole carbon source and continue to grow until the ethanol is also depleted at approximately $t = 6$ h. Throughout both phases, dissolved CO₂ is continuously removed from the liquid by the aeration flow in the form of gas bubbles.

Online measurements of ethanol concentration in the exhaust gas are acquired every five minutes via the gas sensor array. These measurements serve as the sole real-time input to the Digital shadow. Biomass and glucose concentrations are available only through offline sampling — manual extraction of liquid samples followed by laboratory analysis — which provides accurate but infrequent reference values used for validation purposes only.

Re-producing the Results

[YPH21] has provided a source-code of their MatLab implementation, but unfortunately the real Ethanol measurements as well as the real concentration value, measured offline in the laboratory, to evaluate the EKF’s estimations, were not included. However the illustration of the result was available, therefore it was decided to use the same model and simulate the real data (offline measurement) then sample the ethanol each five minutes and add an standard noise and use it as the ethanol measurement then compare the result to the illustration from their work.

Implementation in Python

The reference MATLAB implementation of Yousefi-Darani, Paquet-Durand, and Hitzmann [YPH21] was re-implemented in Python [Pyt23], chosen for its open-source availability, its convenience for rapid prototyping, and prior personal experience with the language. The continuous parts of the model — both the process state and the estimation-error covariance — are integrated with the `solve_ivp` routine of the SciPy library [Vir+20], which numerically solves an initial-value problem $\dot{\mathbf{x}} = f(t, \mathbf{x})$ from a given starting state over a chosen time span and returns the solution at the requested time points.

By default, `solve_ivp` uses the explicit Runge–Kutta method of order 5(4), RK45, which is the Python counterpart of the MATLAB `ode45` solver used in the reference work. RK45 is a single-step, explicit method: to advance one step it evaluates the derivative f at several intermediate points (stages) within the interval and combines them as a weighted average to estimate the new state. The “5(4)” denotes an *embedded Dormand–Prince pair* [DP80]: from the same stage evaluations the method forms

two estimates of the next state, one of fourth and one of fifth order. The difference between them provides an estimate of the local truncation error, which in turn drives *adaptive step-size control* — when the estimated error exceeds the tolerance the step is rejected and reduced, and when it lies comfortably below the tolerance the step is enlarged. The accuracy of the integration is governed by the relative and absolute tolerances, `rtol` and `atol`.

Two implementation details follow from the properties of this solver. First, `RK45` is an explicit method and can become inefficient or even unstable on stiff systems. For this reason the deterministic ground-truth simulator, which must produce a highly accurate reference trajectory, is integrated with the `LSODA` method — which automatically switches between a non-stiff Adams method and a stiff backward-differentiation formula — at tight tolerances ($\text{rtol} \approx 10^{-7}$), whereas the EKF covariance propagation uses the cheaper `RK45`. Second, in the generalisation study (Chapter 2.4) the EKF is instead propagated with a *fixed-step* fourth-order Runge–Kutta scheme (`RK4`) rather than `solve_ivp`. This is a deliberate, non-adaptive choice made for performance: the roughly 860 adaptive-solver calls required per run made `solve_ivp` prohibitively slow, while a fixed-step `RK4` with a small number of sub-steps per measurement interval produced numerically equivalent results in a fraction of the time. The rest of the code contained mostly typical Python programming, the initial values and the equations from the paper were implemented as modular elements. The result, shown in Figure 2.2, matches the result from [YPH21].

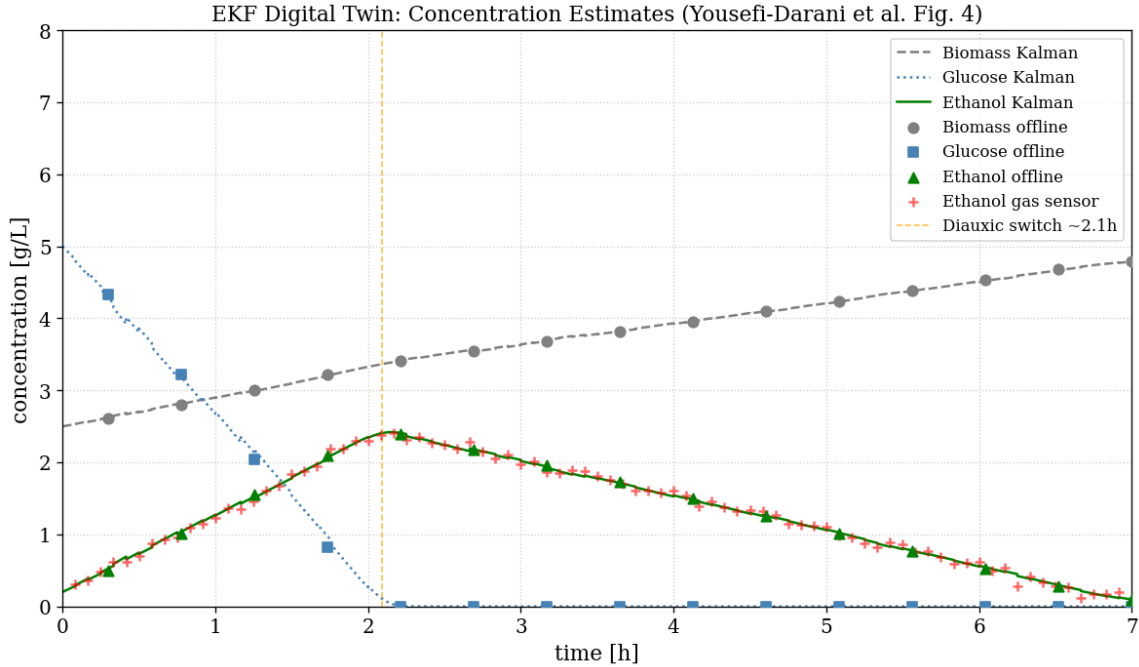


Figure 2.2: Reproduced Final concentration trajectories from [YPH21]

Python Implementation.

The baseline 5-state EKF implementation is available in the Git Repository.

Hypotheses

As mentioned before, this work extends the work of [YPH21] in several directions. One of them is the testing of a set of hypotheses that give a better understanding of the whole system as well as of its limitations and boundaries. Three hypotheses are investigated: the robustness of the estimator to increasing measurement noise (Hypothesis 1), its robustness to sensor drift and growing noise (Hypothesis 2), and the observability limit of the single-measurement filter (Hypothesis 3). Hypotheses 1 and 2 together characterize the estimator under sensor degradation; Hypothesis 3 asks whether an additional, unmeasured state can be recovered from the ethanol sensor alone.

Hypothesis 1 — Sensor Noise Robustness

This hypothesis investigates the robustness of the Extended Kalman Filter to increasing random measurement noise on the ethanol sensor, and identifies the conditions under which the filter maintains acceptable estimation accuracy. It is expected that the filter with the calibrated R performs better compared to the miscalibrated filter.

Experimental setup.

The ethanol sensor noise was modelled as additive Gaussian noise:

$$z_k = E_{true}(t_k) + \epsilon_k, \quad \epsilon_k \sim \mathcal{N}(0, \sigma^2) \quad (2.19)$$

where σ is the noise standard deviation. Two scenarios were evaluated across seven noise levels $\sigma \in \{0.01, 0.05, 0.1, 0.2, 0.5, 1.0, 2.0\}$ g/L:

- **Fixed R :** The measurement noise covariance $R = 0.1$ remained constant regardless of the actual noise level — simulating a miscalibrated filter operating under unknown sensor degradation.
- **Matched R :** $R = \sigma^2$ was updated at each noise level to match the true sensor noise variance — simulating a well-calibrated filter with prior knowledge of sensor quality.

The baseline noise level was $\sigma = 0.05$ g/L. The threshold $\sqrt{R_{fixed}} = 0.316$ g/L marks the point at which the true noise standard deviation exceeds the filter's assumed value.

Results.

Figure 2.3 shows the root mean squared error of prediction (RMSEP) for biomass, glucose and ethanol as a function of the sensor-noise level σ , for both the fixed- R and matched- R configurations.

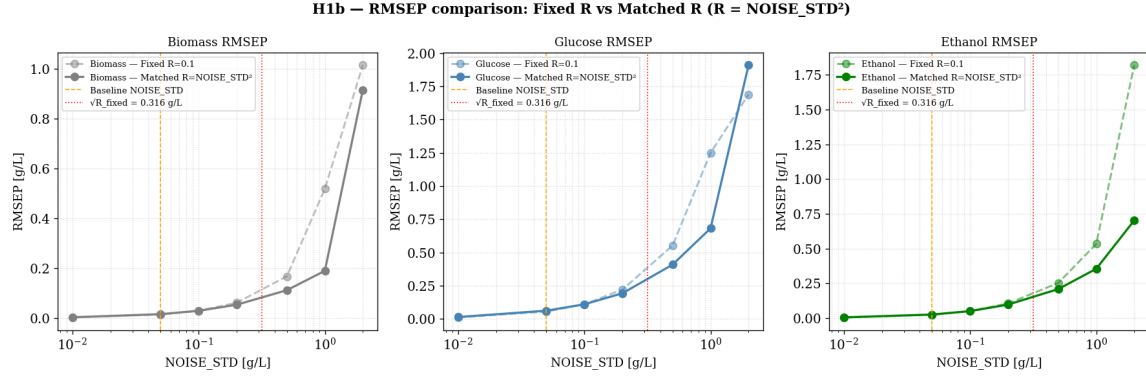


Figure 2.3: Hypothesis 1 — RMSEP for biomass, glucose and ethanol as a function of sensor noise level under fixed R and matched $R = \sigma^2$ conditions. The red dotted line marks $\sqrt{R_{\text{fixed}}} = 0.316$ g/L.

At low noise the two configurations are indistinguishable. For $\sigma \leq 0.2$ g/L all three estimates remain accurate (RMSEP < 0.2 g/L for biomass and ethanol, ≈ 0.19 g/L for glucose), and the fixed- R and matched- R curves lie on top of each other. This is expected: as long as the true noise stays below the assumed measurement standard deviation $\sqrt{R_{\text{fixed}}} = 0.316$ g/L (red dotted line), the fixed filter over-estimates the noise and its Kalman gain remains conservative.

Once σ exceeds $\sqrt{R_{\text{fixed}}}$ the two configurations diverge. Under fixed R the filter now under-models the true measurement noise, so the Kalman gain stays too large and the filter over-trusts the noisy ethanol sensor. Glucose is the most affected across the whole range: because its estimate is driven almost entirely by ethanol through the carbon balance, sensor noise propagates directly into the glucose state, making it the most sensitive of the three.

Matching the filter to the true noise ($R = \sigma^2$) removes a part of this excess error, confirming that the divergence under fixed R is caused by the mismatch between the assumed and the true noise level rather than by the noise itself. The growth is now smoother.

However correct calibration *mitigates but does not eliminate* the fundamental sensitivity of an indirectly estimated state: when the ethanol sensor is this noisy, even a correctly matched R cannot reconstruct glucose from ethanol alone. In practice this means that if the noise level is known the filter should be tuned to it, but for a sufficiently noisy sensor calibration by itself is not enough and a more direct measurement (or an adaptive filtering scheme) is required.

Kalman gain analysis.

The Kalman gain for biomass $K[0]$ and ethanol $K[2]$ across noise levels was calculated. Under fixed R , both gains remained approximately constant — the filter could not adapt its measurement trust as noise increased, because R was not updated. Under matched R , both gains decreased monotonically with increasing noise, demonstrating the theoretically correct behavior: the filter progressively shifted trust from the measurements toward the process model as sensor quality degraded.

Mechanism of collapse.

When the true sensor noise exceeds $\sqrt{R}$, the Kalman gain remains larger than appropriate. Each correction step pulls the state estimates based on unreliable measurements, distorting the covariance matrix P . This distortion propagates to the next Kalman gain computation, producing increasingly erratic corrections — a cascading numerical breakdown. Once P loses its positive definite property, the gain formula produces meaningless values and the filter diverges. Explicitly setting $R = \sigma^2$ prevents this by keeping K appropriately small, so that the filter relies primarily on the model when measurements are unreliable. A DS is supposed to be fed with data from the physical system, specially here where the only connection to the real world is the ethanol's measurement it is then expected that the mechanism collapse when the data does not match the reality.

Hypothesis 2 — Sensor Drift and Growing Noise

Sensor drift and growing sensor noise are two physically distinct degradation modes, and they differ in *which statistical moment* of the measurement error deteriorates. Drift is a systematic, one-directional *bias* that grows over time — for example due to electrode fouling or calibration loss — so the *mean* of the measurement error becomes nonzero and grows. Growing noise is increasing random *imprecision*: the noise *variance* grows over time while the mean error stays zero. This distinction is decisive, because the filter's measurement-noise covariance R can only describe zero-mean random spread — it is a *variance* model. It can therefore correct a growing-variance fault but has no leverage on a growing-mean bias. The following experiments evaluate both modes.

Directional drift.

The drifting measurement was modelled as:

$$z_k = E_{true}(t_k) + \epsilon_k + d \cdot t_k \quad (2.20)$$

where d is the drift rate in g/L/h and t_k is the current time. Two strategies were evaluated across six drift rates $d \in \{0.0, 0.01, 0.05, 0.1, 0.2, 0.5\}$ g/L/h:

- **Unknown drift, fixed R:** The filter had no knowledge of the drift. $R = 0.1$ remained constant. The drift accumulated silently in the measurements.

- **Known drift, time-varying R :** The drift was known but modelled as increased measurement uncertainty: $R(t) = (\sigma_0 + d \cdot t)^2$.

Strategy 1 showed linear RMSEP degradation proportional to the drift rate, with glucose reaching 0.65 g/L at $d = 0.5$ g/L/h. Strategy 2 performed worse than the uninformed baseline at every drift rate: inflating R merely makes the filter trust the biased reading less and lean harder on its model, without removing the bias, so measurement information is discarded for no benefit. This confirms that adjusting R is not an appropriate response to a directional systematic error.

Growing noise.

The sensor with growing noise was modelled as:

$$z_k = E_{true}(t_k) + \epsilon_k, \quad \epsilon_k \sim \mathcal{N}(0, (\sigma_0 + \alpha \cdot t_k)^2) \quad (2.21)$$

where α is the noise growth rate in g/L/h. Two strategies were compared across six growth rates $\alpha \in \{0.0, 0.01, 0.05, 0.1, 0.2, 0.5\}$ g/L/h:

- **Uninformed:** Fixed $R = \sigma_0^2 = 0.0025$.
- **Informed:** Time-varying $R(t) = (\sigma_0 + \alpha \cdot t_k)^2$, matching the true growing noise variance.

Results.

Figure 2.4 shows the root mean squared error of prediction (RMSEP) for biomass, glucose and ethanol as a function of the sensor-noise level σ , for both the fixed- R and Time-varying- R configurations.

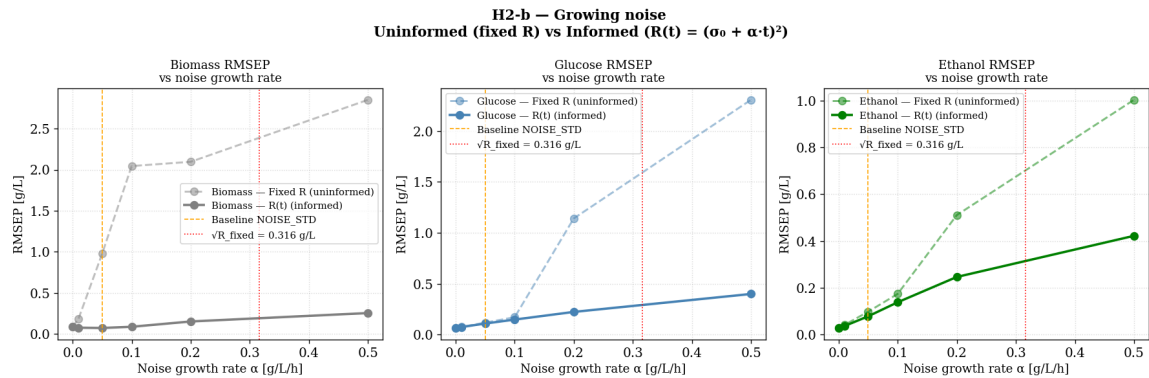


Figure 2.4: Hypothesis 2 — RMSEP for biomass, glucose and ethanol as a function of sensor noise level under fixed and Time-varying R conditions. The red dotted line marks $\sqrt{R_{\text{fixed}}} = 0.316$ g/L.

The uninformed filter degraded rapidly, resembling the Hypothesis 1 collapse pattern. The informed filter with time-varying $R(t)$ performed substantially better across

all variables, with biomass RMSEP remaining below 0.5 g/L even at the highest growth rate. Here $R(t)$ is effective precisely because the degradation *is* a growing variance — exactly what R represents — the opposite conclusion from directional drift.

Finding (Hypothesis 2).

The appropriate mitigation depends on which moment of the error degrades. Growing random noise is a second-moment (variance) fault and is correctly handled by a time-varying $R(t)$ that tracks the actual noise variance. Directional drift is a first-moment (mean) fault and cannot be addressed by modifying R at all: explicit drift compensation is the correct approach when the drift rate is known, and residual-trend (nonzero-mean) analysis enables detection when it is unknown. Misidentifying one mode as the other actively worsens estimation performance.

Summary of Hypotheses 1 and 2 (sensor degradation).

The overarching finding of Hypotheses 1 and 2 is that the EKF is sensitive to the calibration of R — not because of a fundamental algorithmic limitation, but because R encodes the filter’s trust model for the sensor. Accurate characterisation of the sensor degradation mode allows the filter to maintain near-baseline performance under significant degradation. This behaviour is consistent with the filter consistency principle described by Simon [Sim06], who states that deviations in the mean and covariance of the innovation sequence from their theoretically expected values indicate either incorrect noise statistics or a change in the underlying process. The results confirm both sides of this principle: growing or miscalibrated noise manifests as inflation of the innovation *covariance* — a second-moment signature that R can correct — while unmodelled drift produces a nonzero innovation *mean* — a first-moment signature that R cannot — each a distinct and detectable feature of the residual sequence.

Hypothesis 3 — Observability Limit: Dissolved CO₂

This hypothesis investigates the observability limit of the single-measurement EKF framework. The central question is whether an additional unmeasured state variable can be estimated from the ethanol gas sensor alone. As it was mentioned in 2.1.7 Dissolved carbon dioxide (CO₂) is a side product of the *S. cerevisiae*’s cultivation, thus it was chosen as the test variable: unlike a variable that stays essentially constant, its concentration varies markedly throughout the batch, so it is the more demanding test of observability. The state vector was extended from five to six variables and the EKF was run without any additional sensor; the ethanol measurement every five minutes remained the only information channel.

Biological background.

S. cerevisiae produces CO₂ as a metabolic waste product from two pathways. During the glucose phase (Crabtree effect), glucose is simultaneously converted to biomass,

ethanol and CO₂. During the ethanol phase, aerobic oxidation of ethanol produces biomass and CO₂. In both phases, dissolved CO₂ is removed from the liquid by aeration stripping, modelled as a first-order process.

Chen [Che76] demonstrated in submerged yeast culture that growth inhibition was negligible at dissolved CO₂ concentrations below 1.19×10^{-2} M (0.52 g/L), with significant inhibition only observed above 1.6×10^{-2} M (0.70 g/L). Since exhaust gas CO₂ in the present lab-scale batch system peaks at approximately 5–6%, the corresponding dissolved CO₂ concentration remains well below the inhibition threshold. Accordingly, CO₂ inhibition of growth kinetics is not included in the model.

Model extension.

The dissolved CO₂ mass balance was modelled as:

$$\frac{d(\text{CO}_2)}{dt} = \frac{\mu_G \cdot X}{Y_{G,\text{CO}_2}} + \frac{\mu_E \cdot X}{Y_{E,\text{CO}_2}} - k_{\text{CO}_2} \cdot \text{CO}_2 \quad (2.22)$$

where the first two terms represent CO₂ production from glucose and ethanol metabolism respectively, and the third term represents stripping by aeration (first-order, with equilibrium concentration $\text{CO}_2^* \approx 0$ in contact with ambient air).

Parameter derivation.

The yield coefficients Y_{G,CO_2} and Y_{E,CO_2} were derived from a carbon balance using the existing model yield parameters, without requiring additional literature values.

The carbon content of glucose (MW = 180 g/mol, 6 carbons) gives a carbon fraction of $f_C(\text{glucose}) = 72/180 = 0.400$ g C/g. Carbon is distributed to biomass (carbon fraction 0.488 g C/g dry weight [Roe83]), ethanol ($f_C = 0.522$ g C/g), and CO₂ ($f_C = 12/44 = 0.273$ g C/g):

$$\begin{aligned} C_{\text{to CO}_2}^G &= 0.400 - Y_{gx} \cdot 0.488 - Y_{ge} \cdot 0.522 \\ &= 0.400 - (0.15)(0.488) - (0.34)(0.522) = 0.1493 \text{ g C/g glucose} \end{aligned} \quad (2.23)$$

$$Y_{G,\text{CO}_2} = \frac{0.1493}{0.273} = 0.547 \text{ g CO}_2/\text{g glucose} \quad (2.24)$$

Similarly for the ethanol phase:

$$\begin{aligned} C_{\text{to CO}_2}^E &= 0.522 - Y_{ex} \cdot 0.488 \\ &= 0.522 - (0.43)(0.488) = 0.3122 \text{ g C/g ethanol} \end{aligned} \quad (2.25)$$

$$Y_{E,\text{CO}_2} = \frac{0.3122}{0.273} = 1.144 \text{ g CO}_2/\text{g ethanol} \quad (2.26)$$

Both values are physically consistent: $Y_{G,\text{CO}_2} = 0.547$ lies between pure Crabtree fermentation (≈ 0.489) and pure aerobic oxidation (≈ 1.467), and $Y_{E,\text{CO}_2} = 1.144$ lies below pure aerobic ethanol oxidation (≈ 1.913).

The CO₂ stripping rate was estimated from the oxygen mass-transfer coefficient via Higbie penetration theory. The oxygen transfer coefficient $k_L a \approx 44.8 \text{ h}^{-1}$ is derived from the physics of the reactor (agitation and aeration), as detailed in Section 2.5.3:

$$k_{\text{CO}_2} = k_L a_{O_2} \sqrt{\frac{D_{\text{CO}_2}}{D_{O_2}}} = 44.8 \sqrt{\frac{2.0}{2.5}} \approx 40 \text{ h}^{-1}. \quad (2.27)$$

The initial dissolved CO₂ concentration was set to $\text{CO}_{2,0} = 0.05 \text{ g/L}$, reflecting a pre-aerated medium close to equilibrium with ambient air.

Results.

The EKF tracked the CO₂ trajectory with $\text{RMSEP} = 0.002 \text{ g/L}$; however, three independent diagnostic analyses confirmed that this performance was model-driven rather than measurement-driven:

1. **Kalman gain decay.** The Kalman gain for CO₂, $|K[\text{CO}_2]|$, decreased from 0.18 at the start of the experiment to 10^{-4} by $t = 7 \text{ h}$ (five orders of magnitude on a logarithmic scale), indicating that the ethanol measurements provided negligible correction to the CO₂ estimate by the second half of the batch.
2. **Covariance analysis.** The diagonal element $P[5, 5]$ (CO₂ uncertainty) was visually identical before and after each Kalman update throughout the experiment. The uncertainty reduction per update was essentially zero after $t \approx 0.3 \text{ h}$. Furthermore, the cross-covariance $|P[2, 5]|$ between ethanol and CO₂ decayed from 0.04 to zero within the first hour, confirming that the two variables became statistically decoupled and that the ethanol measurement carried no information about CO₂ dynamics beyond the initial transient.
3. **EKF versus pure ODE integration.** Running a pure ODE integration of the CO₂ equation (without any Kalman corrections) produced identical results to the full EKF: both achieved $\text{RMSEP} = 0.002 \text{ g/L}$ and the difference between them oscillated within $\pm 0.004 \text{ g/L}$ only during the first hour before converging to exactly zero. This definitively confirms that the EKF was not estimating CO₂ — it was integrating the model forward without any effective measurement feedback.

Conclusion.

As shown in Figure 2.5; dissolved CO₂ is not observable from the ethanol gas sensor alone. Although the CO₂ signal varies throughout the experiment (the two-peak profile above), the coupling between the ethanol measurement and the CO₂ state is too weak: the cross-covariance $P[2, 5]$ decays to zero within the first hour, and the Kalman gain falls to 10^{-4} by the end of the batch. Adding CO₂ to the state vector neither improves nor degrades the estimation of the original five variables.

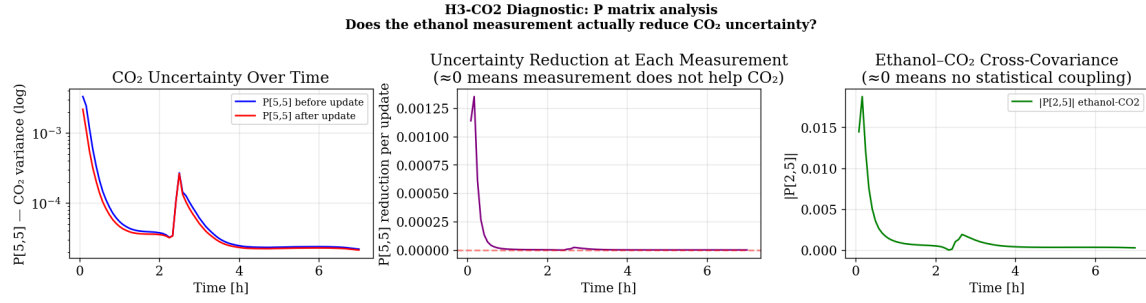


Figure 2.5: Hypothesis 3 — weak coupling between CO₂ and ethanol .

Summary of Hypothesis 3.

The non-observability of CO₂ is particularly informative because it demonstrates that variability in an unmeasured state is a *necessary but not sufficient* condition for observability. The signal must also maintain a persistent coupling to the available measurement — quantified here by the cross-covariance $P[2, 5]$ — which in this case decayed to zero within the first hour of the batch. These findings directly motivate a subsequent investigation in which an exhaust-gas CO₂ sensor is introduced as a second measurement alongside ethanol.

Python Implementation.

The hypotheses testing is implemented in the Git Repository.

Generalization

Since this work aims to provide a generalized roadmap, it is worth investigating how well the baseline EKF performs on other systems whose dynamics are unknown to it. To this end, 25 papers were reviewed, each reporting a bioreactor cultivation of a micro-organism with ethanol and glucose tracked over time. Of these, nine reported the concentration trajectories clearly, together with the governing rate equations for each term and all initial values; these nine were adopted as the reference cases.

Each reference case was reconstructed using the same simulation procedure described in Section 2.2.1. Alternative approaches for generating the ground-truth trajectory were also considered, but the adopted method is identical to that procedure. This choice was motivated by the high degree of internal control it affords over the simulator—a property essential for the subsequent fault-injection step, in which not only healthy batches but also faulty ones must be generated in a controlled manner. No measurement noise was added at this stage, since these trajectories are treated as the noise-free ground truth against which the estimator is evaluated.

Because this reconstruction reproduces each paper’s published *model* rather than digitizing its figures, the resulting trajectories are faithful reconstructions rather than measured data. Where a paper reported only some of the states, the remaining states were generated from the fitted model; consequently a small number of reconstructed

trajectories—most notably some biomass curves—may differ in detail from the corresponding figures in the original publications. As an illustration, the reproduced result for Wang et al. [Wan+04] is shown in Figure 2.6 (reproducing its Fig. 2, Fig. 4 and Fig. 3, in that order).

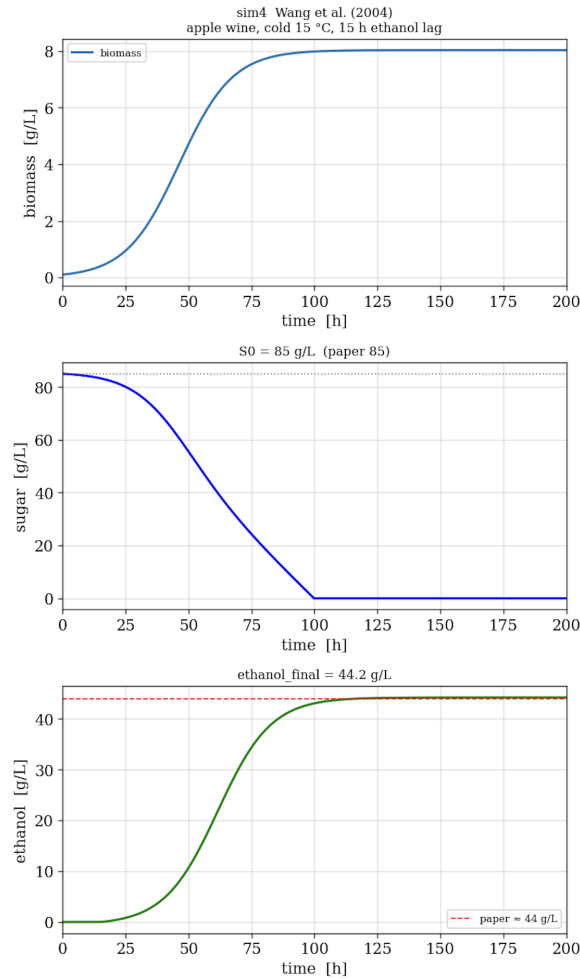


Figure 2.6: Reproduced Final concentration trajectories from [Wan+04]

Selection of the Simulator

Several candidate tools were considered for generating the reference trajectories. The selection was guided by four criteria: the simulator should describe the aerobic batch cultivation of *Saccharomyces cerevisiae* with glucose and ethanol metabolism; it should allow the analyst to set initial conditions and observe the resulting trajectories; and it should impose minimal setup and maintenance overhead so that effort remains focused on the estimation and fault-detection objectives of the thesis.

A survey of available options revealed that no standalone, actively maintained simulator exists that is dedicated specifically to *S. cerevisiae* batch fermentation in a form that can be invoked as a simulator. Most mechanistic fermentation simulators

in the literature are either implemented in MATLAB, target a different organism or cultivation mode, or are themselves ODE-solving frameworks that require the analyst to supply the governing equations—thereby reintroducing the circularity described in 2.2.1. Two categories of tool were examined in detail. The first category comprises object-oriented modelling frameworks such as pyFOOMB [Hem+21], which provide a numerical integrator and parameter management but no built-in biology; using such a framework would amount to re-solving the analyst’s own equations under a different wrapper. The second category comprises web-accessible model repositories, which host curated kinetic models that can be executed remotely and whose results can be exported without any local installation.

On this basis, JWS Online [Pet+17] was tested as the simulation environment. JWS Online is a web-based repository of curated kinetic models that are executed on a remote server; the user interacts with the model through a browser, sets parameter values and initial conditions, runs a time-course simulation, and exports the resulting trajectories in tabular form. Two properties make this environment well suited to the present work. First, the numerical computation is performed remotely, by software the analyst neither wrote nor maintains, which makes the resulting data genuinely independent of the estimator implementation. Second, the export of trajectories as comma-separated values provides a clean, language-agnostic interface: the reference data can be consumed by the estimation pipeline without any coupling between the two environments. After running the simulation using JWS-Online [Pet+17] the trajectories were generated as shown in Figure 2.7 Considering the future needs of

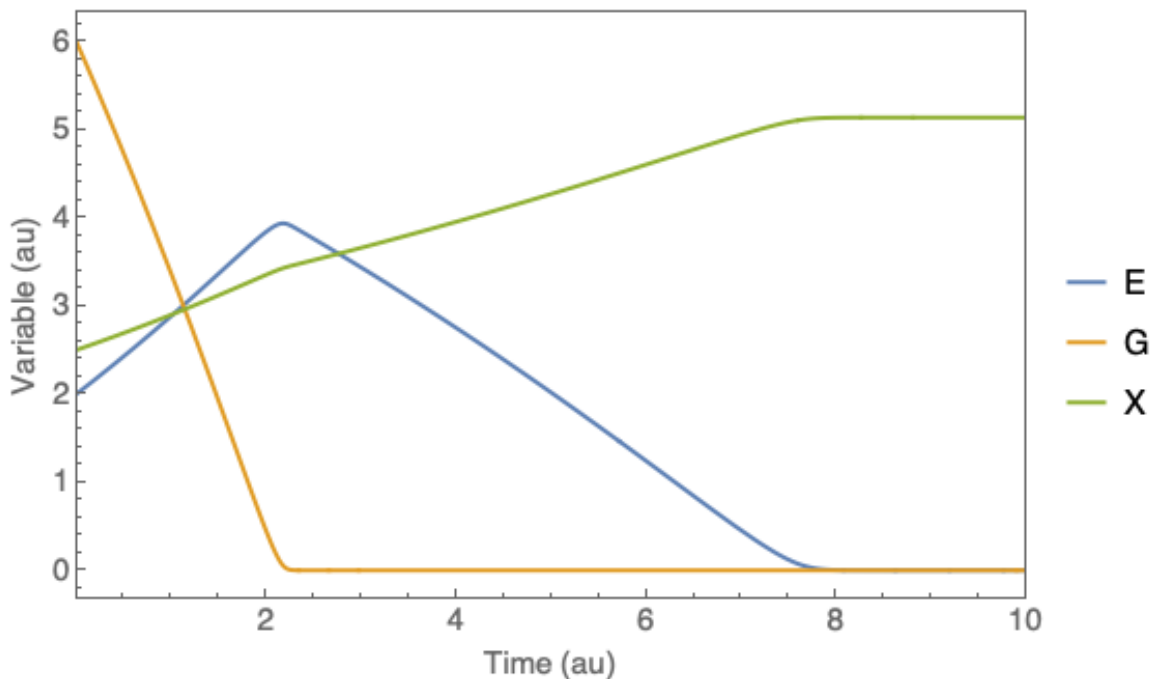


Figure 2.7: Generated Concentration Trajectory.

deliberate fault injection in the middle of the simulation and the lack of control in the middle of the cultivation in JWS, the final decision was not in JWS’s favor,

however the advantages of this framework was a great motivation for a web-based user-interface, where different experiments can be initialized easily and the results are clear.

UI Implementation.

Up to this point, every run and experiment was carried out in a Python environment, so that each new scenario required editing the code — a time-consuming process whose results were, moreover, accessible only from within the development environment. To make the results clearer and the underlying implementation usable also by users without programming experience, such as biologists and chemists, and to allow different scenarios to be run without changing any code, a graphical user interface was built on top of the existing Python implementation.

The interface follows a client–server architecture. The backend is a Python application built on the FastAPI web framework [Ram18] and served by the Uvicorn ASGI server; it exposes the digital shadow through a REST endpoint that supplies the default parameters and a WebSocket that streams the live simulation step by step. It reuses the same scientific stack as the rest of this work — NumPy [Har+20], SciPy [Vir+20] for the ODE integration, SymPy [Meu+17] for the symbolic EKF Jacobian, and Matplotlib [Hun07] for server-side figure export. The physics and estimator modules deliberately contain no interface code, so they can equally be driven from a script or a notebook. The frontend is a lightweight browser page (plain HTML and JavaScript, with no build step) that renders the live state interactively using the Plotly graphing library [Plo15]. The interface exposes all initial conditions and the cardinal temperatures and pH as editable inputs, and is structured so that further bioreactors and organisms can be added and selected without changing the frontend.

The UI can be accessed in the Bioreactor UI Documentation. It can be easily used by following the instruction in README.md in the same folder.

Reproduction of the Nine Reference Models

To summarize the preceding steps: there was a clear motivation to examine how the designed EKF behaves when reality differs from the model on which it relies. Here, “differs” does not denote a fundamentally different cultivation—in every case the process is still a yeast growing on sugar and producing ethanol—but rather the same type of process realized under the different conditions reported in the nine published studies tested below. The direct use of `solve_ivp` provides full control over the model, which is required both for injecting faults in the subsequent sections and for modifying the Monod kinetics to reproduce the results of the nine papers. `solve_ivp` therefore remained the preferred choice.

As stated earlier, the nine candidates were selected for three reasons: (i) they share the same underlying process—a yeast fermenting sugar (predominantly glucose) into ethanol; (ii) they provide the mathematical formulation of their Monod-type growth kinetics, so that their results are reproducible; and (iii) they report a clear final

trajectory of the concentrations, allowing the results simulated with `solve_ivp` to be compared directly against the published curves.

It should be noted, however, that the nine models are not identical in metabolic scope: only two of them—simulation 8[Sal+23b] and simulation 9[GMP22]—reproduce the diauxic switch in which the yeast, after depleting the sugar, re-consumes the ethanol it had produced. In the remaining seven, the ethanol accumulates monotonically as a purely fermentative product and is never consumed. This distinction is relevant because the baseline EKF is itself built around a diauxic model, so the seven monotonic-ethanol plants already constitute a structural mismatch between the filter’s assumptions and the reference process. Once each reference trajectory had been reproduced, the nine data sets were used as ground truth and the EKF was run on each in turn. As in the baseline model, the ethanol concentration was sampled from the corresponding CSV file every five minutes, so that the resulting RMSEP values remain comparable across all cases. The results for simulation 3[Pin+17] and simulation 4[Wan+04] are shown in Figure 2.8 and Figure 2.9.

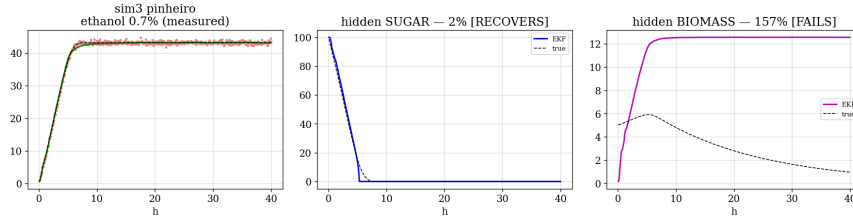


Figure 2.8: EKF results using the model provided [Pin+17]

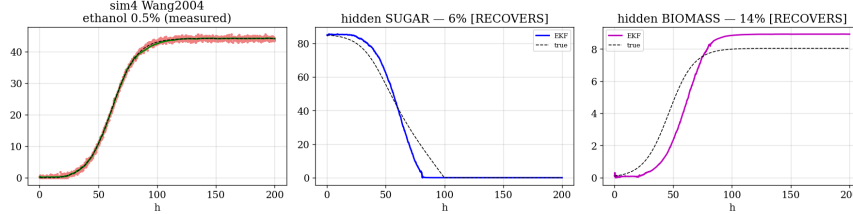


Figure 2.9: EKF results using the model in [Wan+04]

Simulations

The EKF is kept simple: Ethanol measurement every 5 minutes (sampled from ground truth derived from the simulators) using a 5-variable state vector: Biomass, Glucose, Ethanol, maximum growth using glucose and maximum growth rate using ethanol. The important question now is: Is this EKF a generalized filter? The goal of this part is to realize how capable is the EKF of estimating the parameters it has no direct value of. This part is inspired from the core paper [YPH21] where they tested if the EKF can estimate the Glucose and Biomass-concentration using only the Ethanol measurement and here we pursue their work by adding more elements in to the

simulation in order to generalize the experiment. "how far can a simple model track reality when reality has unmodeled factors"?

The first experiment was based on [DBK22]. They developed a glucose-to-ethanol model for immobilized *S. cerevisiae* on sodium alginate beads for Batch glucose fermentation; varied yeast loading, pH, and temperature.

The second one is based on [Sal+23a]. They modeled HG ethanol fermentation of sweet sorghum juice by *S. cerevisiae* NP 01.

The third one is based on [Pin+17]. They studied ethanol production by flocculent *S. cerevisiae* CCA008 on cashew apple juice (Figure 2.8).

The fourth one is based on [Wan+04]. They predicted glucose, fructose, and sucrose use in apple wine yeast fermentation. The result of these experiments are shown in Figure 2.9.

The fifth one is based on [Kon+19]. They developed a non-structured (logistic) model for free-floating *S. cerevisiae* ATCC 7754 fermenting glucose in a 2 L stirred bioreactor—a small, low-gravity batch (biomass ~ 0.34 g/L, ethanol ~ 5 g/L).

The sixth one is based on [Ari+16]. They modelled ethanol production by *S. cerevisiae* NP 01 immobilized on sweet-sorghum stalk pieces, fermenting a very sugary juice (~ 225 g/L) to ~ 100 g/L ethanol, with both growth and production limited by substrate and product inhibition.

The seventh one is based on [Zin+16]. They modelled a large amount of pre-grown, *non-growing* *S. bayanus* acting as a fixed biocatalyst that converts glucose and fructose simultaneously into ethanol, each sugar inhibited by the ethanol produced (fructose more strongly than glucose).

The eighth one is based on [Sal+23b]. They developed a mechanistic model for a different genus, *Kluyveromyces marxianus*, fermenting glucose in a fast batch in which cells die (both naturally and from ethanol toxicity) and ethanol is produced and then slowly re-degraded, so it rises and then falls.

The ninth one is based on [GMP22]. They proposed a comprehensive mechanistic model of *S. cerevisiae* able to switch metabolism with the growth conditions: aerobic (Crabtree) fermentation while glucose is abundant, switching to respiration—first of the residual glucose, then of the ethanol itself—once glucose is depleted.

Results

The results in Figure 2.9 can be regarded as a successful case: using a single measured variable (ethanol) and a deliberately mismatched kinetic model, the EKF still reconstructs the hidden states—sugar and biomass—with acceptable accuracy (6 % and 14 % RMSEP, respectively). The reference process here is the apple-wine fermentation of Wang et al. [Wan+04], in which *S. cerevisiae* grows according to a logistic law at 15 °C with a pronounced (≈ 15 h) lag before ethanol formation; crucially, the biomass increases monotonically towards a stable plateau and no cell death occurs. Why the recovery of the hidden states from a single sensor matters becomes clearer in the root-cause-analysis part, where these estimated internal states are what allow a fault to be localised.

Figure 2.8, by contrast, tells a different story, most visibly in the biomass, where the estimate diverges from the truth ($\text{RMSEP} \approx 157\%$). Inspecting the underlying process model explains this apparent failure: the reference is the cashew-apple-juice fermentation of Pinheiro et al. [Pin+17], which describes a flocculent yeast with an explicit viable/dead-cell distinction, so that the true viable biomass rises to a peak and then declines as cells die. The general EKF used here is built around a growth-only, ethanol-coupled biomass model and contains no death term; it therefore drives the biomass estimate upward to a plateau while the true viable biomass falls, producing the large error. This is compounded by the fact that biomass is not observable from ethanol alone, so the measurement cannot correct the structurally wrong prediction.

It is worth emphasizing that model-based Bayesian estimators such as the Kalman filter and its non-linear extensions do not learn a data-driven mapping in the way that, for example, neural networks do; their accuracy is bounded by the fidelity of the assumed process model. It is therefore unsurprising that a single, general-purpose EKF—one not tailored to a specific experiment—cannot recover a state that is simultaneously unobservable from the available measurement and governed by dynamics absent from its model. Across a broad and varied range of monotonic fermentations (e.g. Figure 2.9) the simple, general EKF performs well; however, when the reference process departs strongly from the filter’s assumptions (e.g. Figure 2.8), additional information—such as biomass sampled online or through laboratory assays—is required to restore accuracy.

Evaluation

Cadence experiment.

To quantify how a direct measurement of a hidden state improves digital-twin accuracy, the nine reference simulators were each driven through the extended Kalman filter under two configurations: a reference configuration matching the original setup—ethanol sampled every five minutes as the sole correction signal—and an augmented configuration that retains the same ethanol stream and adds a second sensor, either glucose or biomass, sampled at one of $\{120, 60, 30, 15, 10\}$ min. Sensor noise was set to 1% of the ethanol peak and 5% of the second sensor’s state normalizer, and each plant–cadence combination was repeated over twelve noise seeds. Table 2.1 reports the mean normalized RMSE across eight simulators; simulation 6[Ari+16] is excluded from this mean as a filter-class mismatch (a growth-coupled filter applied to a non-growing biocatalyst). Its effect is shown separately in the nine-simulator mean of Table 2.2 and explained below.

Why biomass is the hardest state to estimate.

The asymmetry between glucose and biomass in Table 2.1 is not a numerical artifact but a direct consequence of observability. The filter’s only correction signal is ethanol, and glucose is tied to ethanol by a conservation law: to first order the consumed sugar equals the produced ethanol divided by the ethanol-on-sugar yield,

Table 2.1: Mean state-estimation error across the eight reference simulators. Ethanol is sampled every 5 min on every plant; the second-sensor cadence is varied. Sensor noise: 1 % of the ethanol peak and 5 % of the state normalizer for the second sensor. Each cell is the mean over twelve noise seeds, then averaged across the eight simulators.

sampling cadence	glucose NRMSE	biomass NRMSE
ethanol only	14 %	99 %
every 120 min	7 %	32 %
every 60 min	6 %	25 %
every 30 min	6 %	19 %
every 15 min	5 %	15 %
every 10 min	5 %	12 %

$S(t) \approx S_0 - E(t)/Y_{GE}$. Every ethanol measurement therefore also constrains glucose, which is why the glucose error remains near 14 % even with no glucose sensor at all. Biomass possesses no comparable link to ethanol: depending on organism and regime, a given amount of ethanol can be produced with widely differing amounts of biomass—the ethanol-on-biomass yield varies several-fold across the reference plants, and in a non-growing biocatalyst ethanol is produced at essentially constant biomass. Consequently the ethanol–biomass cross-covariance decays, the Kalman gain for biomass carries no useful correction, and the estimate reduces to the open-loop model prediction $X_0 + E(t)/Y_{EX}$ with an assumed yield that is itself subject to model mismatch. The difficulty is compounded in plants with a death phase or growth-decoupled kinetics, where the true biomass is not even monotonic and the growth-coupled filter is structurally unable to follow it.

The 6th simulation outlier: a model-applicability limit.

The large biomass error in Table 2.2 originates almost entirely from a single plant, and reflects a structural mismatch rather than a sampling deficiency. The 6th simulation is a *fixed-biocatalyst* fermentation: a large amount of pre-grown, non-growing yeast (≈ 6.2 g/L) converts a very high sugar load (≈ 400 g/L of glucose and fructose) into ethanol (up to ≈ 182 g/L) over 250 h, while the biomass itself remains essentially constant. This violates the model’s core assumption—that metabolism and growth are coupled—and is not a limitation of the estimator’s tuning or of the sampling cadence: a direct biomass probe reduces the error (from 740 % to about 69 % at a fifteen-minute cadence) but cannot fully repair a structurally inapplicable model, because the process model keeps over-predicting growth between samples. Simulation 6 is therefore retained in Table 2.2 for transparency but excluded from the eight-simulator mean, where it would otherwise inflate the biomass column by roughly seventy percentage points without informing the sampling question.

Table 2.2: Mean state-estimation error across all nine reference simulators. The growth-coupled filter mismatch on simulation 6 produces a 740 % baseline biomass error, lifting the ensemble mean by approximately seventy percentage points; this is a filter-class mismatch, not a sampling problem.

sampling cadence	glucose NRMSE	biomass NRMSE
ethanol only	14 %	170 %
every 120 min	7 %	36 %
every 60 min	6 %	28 %
every 30 min	5 %	21 %
every 15 min	5 %	15 %
every 10 min	5 %	12 %

Operational choice.

Both probes reduce the error monotonically with cadence, and the marginal benefit flattens below 60 min. For applications where accuracy is the dominant concern, hourly sampling of both biomass and glucose is recommended. For the remainder of this study, however, only hourly biomass sampling is retained: glucose is already well constrained by ethanol through the carbon balance on most plants (14 % mean baseline, 6 % at 60 min) and the value of an additional glucose sensor is plant-specific, whereas biomass has no conservation anchor to ethanol (99 % mean baseline, 25 % at 60 min) and a dedicated probe is therefore required universally. Tighter cadences remain available where higher accuracy is needed: 15 min biomass sampling brings the biomass error to 15 % (Table 2.1). The effect of the two added sensors can also be seen directly in the estimated trajectories: Figure 2.10 shows the biomass estimate with an hourly biomass probe, and Figure 2.11 the glucose estimate with an hourly glucose probe. In both cases the filtered curve is pulled onto the reference trajectory that ethanol alone could not constrain, consistent with the error reductions reported in Table 2.1. and 2.11

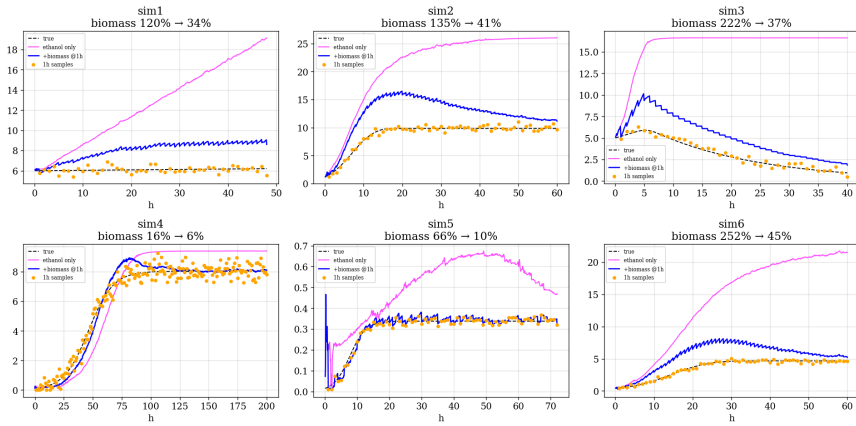


Figure 2.10: Biomass-concentration with vs without the Biomass sample

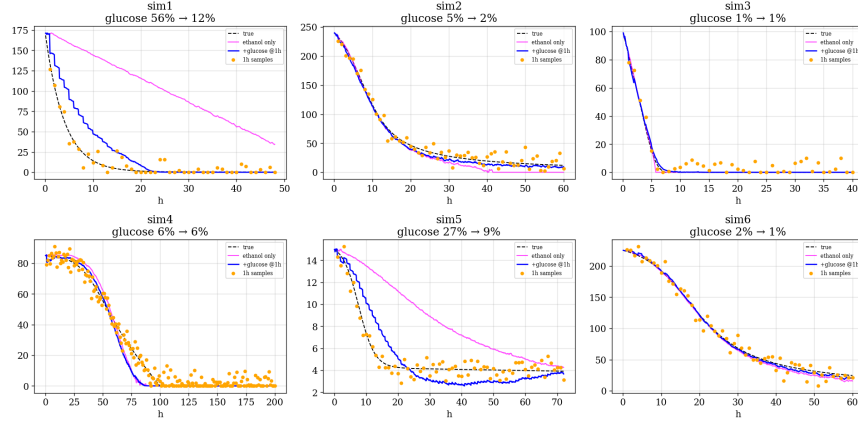


Figure 2.11: Glucose-concentration with vs without the Glucose sample

Python Implementation.

The generalization testing is implemented in the Git Repository.

Fault Detection and Root Cause Analysis

As stated in the abstract, this work comprises two major parts: the development of a digital shadow of a simple bioprocess and the use of that shadow to perform root cause analysis. Although the development part is not yet complete, it is convenient to introduce the second part at this point and to continue both in parallel. A brief review of the standard notion of RCA is therefore given first.

Root cause analysis was originally developed in psychology and systems engineering to identify “the basic and causal factor(s) that underlie variation in performance.”[WLP08]. In one sentence, Root cause analysis (RCA) is the process of identifying the fundamental cause of a problem through a structured procedure [SMG13]. Locating the cause of a failure, however, presupposes that the failure has first been *detected*, and it is here that the value of the developed digital shadow becomes most apparent: the extended Kalman filter at the core of the shadow can serve as the detection mechanism itself. As discussed in Section 2.1.4, at every sampling instant the EKF performs a correction step driven by the *innovation*—the difference between the measured ethanol concentration and the value predicted by the healthy model,

$$\nu_k = z_k - H \hat{x}_k^-, \quad (2.28)$$

where $\hat{x}_k^-$ is the prior state estimate and H is the measurement matrix that extracts ethanol from the state vector. The size of this innovation is judged against its expected spread, the innovation covariance

$$S_k = H P_k^- H^\top + R, \quad (2.29)$$

which combines the propagated state uncertainty P_k^- with the measurement-noise covariance R . In words, ν_k quantifies how far the real measurement departs from

the healthy-model expectation, while S_k encodes how large such a departure may plausibly be, given the model uncertainty—carried by P and, through the prediction step, by the process-noise covariance Q —and the sensor noise carried by R . Under healthy operation the normalized innovation $\nu_k/\sqrt{S_k}$ fluctuates around zero within a bounded band set by the noise; under a fault the filter continues to predict the healthy trajectory, so the discrepancy grows from step to step and the innovation drifts systematically away from zero [VCM12]. Distinguishing this fault-induced growth from the ordinary, noise-bounded fluctuation of a healthy batch is the central detection problem, and the fault scenarios simulated later are used precisely to locate that boundary.

The first task is therefore to establish a practical method for fault detection, followed by a method for root cause analysis. To this end, the relevant literature is reviewed in the following section.

Background Research

To establish a reusable and general road-map for root cause analysis, twenty-five publications were reviewed and seven were combined into the design adopted in this work. The literature was located using search terms including *residual / innovation-based fault detection for Kalman filters*, *fault detection and isolation with the Kalman filter*, *digital-twin-driven fault detection*, and *root cause analysis methods, tool selection, best practice, frameworks and techniques* (including five-whys and cause-effect trees).

Approaches to root cause analysis can be grouped into four broad families, which differ mainly in how much they rely on human expertise, on historical data, and on an explicit process model.

The oldest family consists of the *qualitative, expert-driven* tools of quality engineering: the five-whys technique, the Ishikawa (cause-and-effect) diagram, fault-tree analysis, and failure mode and effects analysis (FMEA) [Wil01; Gan19; SMG13]. These methods provide a structured way to reason from a symptom to a plausible cause and remain standard practice in manufacturing, healthcare [WLP08] and bio-pharmaceutical risk assessment, where FMEA in particular is widely applied to bioprocess development [Mol05; Sta14; Bio26]. Their strength is transparency and the incorporation of domain knowledge; their limitation is that they are manual, qualitative and retrospective, and therefore a poor fit to the continuous, real-time monitoring of a dynamic cultivation.

A second family replaces expert reasoning with *data-driven, multivariate statistics*. In the process and bioprocess industries, root cause analysis is commonly performed with multivariate statistical process control—principal component analysis and partial least squares together with contribution plots—or by comparing a running batch against a “golden” reference batch [Bor+19; Luo+24]. Borchert et al. [Bor+19] show for bio-pharmaceutical processes that raw-data and feature-based workflows each identify the variables responsible for a deviation, and that fusing them yields the most effective analysis, while Luo et al. [Luo+24] apply a golden-batch reference directly to anomalies in bioreactor fermentation. Such methods exploit the large data

historians now available, but they depend on representative historical data, they attribute a deviation to correlated *variables* rather than to a physical *cause*, and they extrapolate poorly to operating regimes not seen during training. Machine learning sharpens the classification step but inherits the same reliance on labeled fault data and offers limited interpretability for faults unseen during training [Hu+25].

A third family is *model-based fault detection and isolation* (FDI), which provides the quantitative backbone on which this work builds. Here an analytical model of the healthy process generates residuals—the discrepancy between measured and predicted behavior—whose statistical signature is then tested for faults. Isermann’s framework organizes this into the stages of detection, isolation and identification and provides the accompanying fault taxonomy [Ise06]. When the estimator is a Kalman filter, the natural residual is the innovation sequence, and monitoring its mean and covariance yields sensitive sensor- and actuator-fault diagnosis [HC00]; adaptive extended and unscented Kalman filters have been used to detect and isolate faults from this innovation stream in aerospace and process applications [VCM12; DSG14]. The decisive advantage over the statistical family is that the residuals are physically meaningful: because they are defined against a first-principles model, a deviation can be attributed to a specific mechanism rather than to a correlated measurement.

Within the model-based family, *multiple-model adaptive estimation* (MMAE) extends residual monitoring from detection to explicit root cause identification. A bank of Kalman filters is run in parallel, each embedding a different hypothesis about the system, and a Bayesian recursion over their innovation sequences assigns a posterior probability to each hypothesis, so that the model best explaining the measurements identifies the active fault [HM00]. This hypothesis-testing view maps directly onto root cause analysis: each candidate cause is encoded as a model, and the data themselves select the most probable one. It is precisely this mechanism that the present work adopts, with a bank of extended Kalman filters—each carrying a candidate fault—competing to explain a single ethanol measurement.

In summary, qualitative tools are transparent but manual and retrospective; statistical and machine-learning methods scale with data but require historical fault records and identify variables rather than causes; and model-based FDI produces physically interpretable residuals but has been applied to bioreactors far less than to mechanical or aerospace systems. The gap this work addresses lies at the intersection of the last two families: a digital shadow of a bioreactor that generates a physically meaningful residual from a single measurement, and that turns that residual into a root cause by means of a bank of hypothesis filters—combining the interpretability of the model-based approach with the automation of a data-driven monitor.

Methodology

Given the constraints of this work—limited expertise in the biological aspects of the process and the absence of real cultivation data for legal and privacy reasons—and building on the methods reviewed above, the following pipeline was proposed and implemented:

1. **Functional diagram.** A functional diagram of the system (Figure 2.12) is constructed to make explicit how the individual components—process model, estimator and measurement—interact and influence one another.
2. **Fault definition.** The set of plausible faults is enumerated, that is, the question “*what can realistically go wrong in this environment?*” is answered for each actuator and subsystem (heater, pH pumps, aeration/agitation and the sensors).
3. **Fault injection and thresholding.** Each fault is injected into the ground-truth simulator and its effect—particularly on the EKF innovation—is documented, in order to establish a threshold that separates a genuine fault from the ordinary, noise-bounded drift inherent to the process.
4. **Fault detection.** A fault-detection function is implemented on the basis of this threshold, monitoring the innovation sequence online.
5. **Root cause analysis.** Finally, a bank of extended Kalman filters is constructed, each embedding a single fault hypothesis; the hypothesis that best explains the measurements identifies the root cause.

Limitation and challenges

1. Lack of expertise in the second phase - a team member who knows what can go wrong in a bioreactor
2. All the available data are simulated - lack of real data may affect the results
3. Many to one relationship - both pH and Temperature failure affect the growth rate - may cause ambiguity

Solutions

1. Find more failure cases in papers - use other’s experiment in lab to find the potential special failure cases.
2. Data-augmentation - we have 10 different simulations with their results. We can change the sensor noise, add drift, change the initial parameters or initial pH to have more data.
3. The ambiguity can be potentially resolved probabilistically through the *likelihood* of the competing hypotheses: each candidate cause is scored by how well its model explains the measurements, and normalizing these likelihoods (Bayes’ rule) yields a probability distribution over the causes — for example, a fall in the growth rate reported as 60% pH too low, 35% temperature too low, and 5% a false alarm.

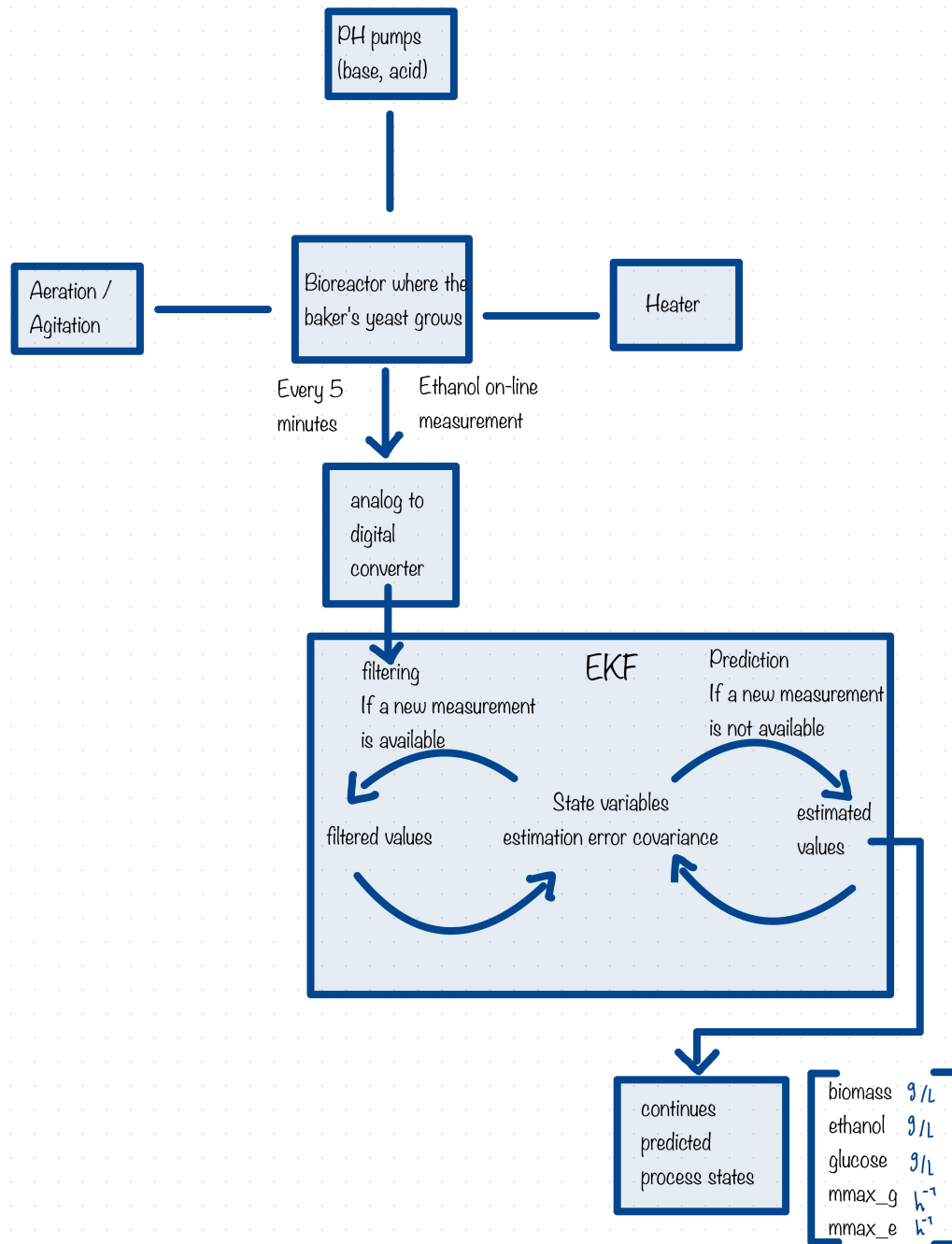


Figure 2.12: The Function Diagram: Extended Kalman filter as a digital shadow of the baker's-yeast bioreactor: the ethanol measurement drives the predict-update cycle, whose state estimate forms the shadow and whose innovation feeds fault detection and root cause analysis. inspired by Fig.3 in [YPH21]

A deeper dive into the physical system.

To proceed, a more detailed understanding of the physical system is required. As noted in Section 2.1.7, the bioreactor used in work [YPH21] was a Minifors (Infors HT); its data sheet was therefore consulted to characterise the subsystems that the fault model depends on [Inf25]. Two of these warrant particular attention, because they are the actuators later subjected to faults: the heating system and the agitation system, together with the vessel geometry that couples them to the process.

Heating. Temperature is controlled by an electrical thermal block with a nominal heating power of about 630 W (peak electrical consumption ≈ 800 W during the heating phase), covering a range of 0 °C to 145 °C [Inf25]. Heat is saturated at a maximum power, with passive loss to the surroundings, rather than active cooling.

Vessel (walls). The culture vessel is a cylindrical, flat-bottomed borosilicate-glass vessel available in three sizes, with nominal inner diameters of 90 mm (DN90, 1.5 L total), 115 mm (DN115, 3 L) and 145 mm (DN145, 6 L) and corresponding heights of 235, 295 and 370 mm [Inf25]. The inner diameter and the liquid height fix the wall area that governs the heat-loss term $UA(T - T_{\text{amb}})$ and, with the working volume, the thermal inertia $V\rho c_p$ of the energy balance.

pH control (dosing pumps). The pH is regulated by the addition of acid and base through peristaltic dosing pumps over a control range of pH 2 to 12 [Inf25]. The pumps operate either at variable speed or as digital ON/OFF units, with typical delivery rates of the order of a few milliliters per minute (about 1 mL min⁻¹ to 16 mL min⁻¹ with water at maximum speed, depending on the pump head) [Inf25]; in the reference setup the titrants are sulfuric acid (H₂SO₄) and sodium hydroxide (NaOH). These pumps are the actuators of the pH subsystem: in the model of Section 2.5.3 they appear as opposing acid and base flows, each saturating at a maximum dosing rate, that counteract the acidification produced by cell growth. Because the physical pumps are digital ON/OFF devices, the throttled proportional (or PI) titration used in the model is a deliberate simplification, adopted to obtain a smooth, readable pH response for fault injection rather than a bang–bang limit cycle.

Agitation. Mixing and oxygen transfer are provided by (typically two) six-blade Rushton turbine impellers mounted on a single, counter-clockwise shaft (AISI 316L for DN115/DN145, PEEK for DN90), with impeller diameters of 38, 46 and 54 mm for the three vessel sizes and a stirrer motor rated up to 260 W [Inf25]. In the reference operating point the stirrer runs at 500 rpm with an aeration rate of 3.5 L min⁻¹; the impeller diameter D and speed N set the stirrer power ($P_0 \propto N_p \rho N^3 D^5$) and hence the volumetric oxygen-transfer coefficient k_{La} used in the dissolved-oxygen balance through the correlation of [van79].

Fault definition.

Building on the system diagram (Figure 2.12) and the detailed description of the physical system (Section 2.5.2), the set of plausible faults can now be enumerated. To keep this enumeration systematic, the faults that may occur in the combined environment—the digital shadow together with the physical reactor—are grouped

into three categories, in line with the usual separation between design, operation and degradation in fault taxonomy [Ise06]. *Design* faults originate in an incorrect or only partially correct specification of the digital shadow itself—for example mis-estimated bio-kinetic parameters such as yields or maximum growth rates—and manifest as a persistent bias in the innovation rather than as a physical breakdown. *Operational* faults arise when a component of the physical system fails during a run, the most obvious example being the heater, also the titrant pumps or the aeration and agitation system. *Ageing* faults are caused by time and wear, such as the gradual drift or growing imprecision of a sensor. In this work the focus was mainly on the 3 main operation fault caused by heater, titrant pumps and aeration-agitation system, where they lead to an observable impact on the cultivation.

The complete set of considered cases is listed in Table 2.3. It was compiled by the author from the subsystems identified in the physical-system analysis, from domain knowledge acquired during this work, and from fault modes reported for comparable cultivations in the literature. Although the specific faults differ from one experiment to another, the proposed method is general: it applies to any fault that leaves a measurable signature on at least one monitored quantity—either indirectly, through its effect on the ethanol measurement, or directly, when the affected variable is itself measured, in which case the fault is identified without relying on a secondary effect. Conversely, a fault that leaves no measurable signature cannot be detected by any estimator, and two faults producing an identical signature cannot clearly be told apart. Within this scope, the present work concentrates on eight operational cases listed in 2.5.4.

An Extension of the core EKF

Temperature.

This section introduces the thermal layer and the effect of temperature T on growth. Microbial growth exhibits a characteristic temperature dependence: there is an optimum T at which the specific growth rate is maximal, and a range $(T_{\min}, T_{\max})$ outside of which growth ceases; as T approaches either limit the rate falls off. This behavior is captured by the *cardinal-temperature model* of Rosso et al. [Ros+95], which parameterize the response by the three characteristic temperatures $T_{\min}$, T_{opt} and $T_{\max}$; the same growth-boundary approach has since been applied to other organisms, for example by Valero et al. [Val+09]. The cardinal values used here are the *S. cerevisiae* figures of [Sal+11], namely $T_{\min} = 5^\circ\text{C}$, $T_{\text{opt}} = 30^\circ\text{C}$ and $T_{\max} = 40^\circ\text{C}$; the resulting factor $f_T(T)$ is re-anchored so that $f_T(T_{\text{set}}) = 1$ at the operating point.

The vessel is modeled as a perfectly mixed, jacketed stirred tank with an electrical heating element under proportional control. An energy balance on the liquid phase gives

$$\frac{dT}{dt} = \frac{Q_{\text{heat}}(T) - Q_{\text{loss}}(T) + Q_{\text{metab}}}{V \rho C_p}, \quad (2.30)$$

Table 2.3: Fault Classification, Propagation, and Detectability

Category	Fault	Primary propagation
Design	Wrong EKF parameters (Y_{gx} , Y_{ge} , $\mu_{\max}$)	filter $\neq$ reality from $t=0$
Design	Wrong setpoint (T, pH)	constant offset on kinetics
Design	Wrong initial value in EKF	e.g. early/late glucose deplete
Operational	Heater/cooler failure	T excursion $\rightarrow \mu_{\max}$
Operational	Acid/base pump stuck	pH excursion $\rightarrow \mu_{\max}$
Operational	Acid/base pump failure	pH excursion $\rightarrow \mu_{\max}$
Operational	Agitator failure	$kLa \downarrow \rightarrow DO \downarrow \rightarrow \text{growth} \downarrow$
Operational	Aeration	$kLa \downarrow \rightarrow DO \downarrow \rightarrow \text{growth} \downarrow$
Ageing	Sensor drift	wrong measurement
Ageing	Agitation element scaling	$DO \downarrow \rightarrow \text{growth} \downarrow$
Ageing	Aeration element scaling	$DO \downarrow \rightarrow \text{growth} \downarrow$
Ageing	Heater element scaling	T excursion $\rightarrow \mu_{\max}$
Ageing	pH electrodes scaling	wrong measurement

with

$$Q_{\text{loss}}(T) = UA(T - T_{\text{amb}}), \quad (2.31)$$

$$Q_{\text{metab}} = Y_{Q/X}(\mu_G + \mu_E)XV, \quad (2.32)$$

$$Q_{\text{heat}}(T) = \max(0, \min(Q_{\text{max}}, K_p(T_{\text{set}} - T))). \quad (2.33)$$

Here UA is the overall heat-loss coefficient [W K^{-1}]. It follows from the wall construction: for a wall layer of thickness B , thermal conductivity k and area A the conduction resistance is $R = B/(kA)$, the heat flow is $Q = \Delta T/R_{\text{total}}$ across all layers, and the overall coefficient is $UA = 1/R_{\text{total}}$ (equivalently $U = 1/(R_{\text{total}}A)$) [Dor13]. The wetted wall area A of a laboratory vessel of this scale is of the order of 0.04 m^2 to 0.06 m^2 . Representative values for a Minifors-class reactor were adopted; they are listed in Table 2.4.

pH.

As with temperature, microbial growth has a pH range: below a minimum $\text{pH}_{\min}$ and above a maximum $\text{pH}_{\max}$ no growth occurs, with an optimum pH_{opt} in between. Following the same approach as in Section 2.5.3, this dependence is described with

Table 2.4: Model parameters and simulation baseline values (thermal layer).

Parameter	Symbol	Value	Unit	Source
Working volume	V	1.35	L	[YPH21]
Volumetric heat capacity	ρC_p	4184	$\text{J L}^{-1}\text{K}^{-1}$	water heat capacity
Setpoint temperature	T_{set}	30	$^{\circ}\text{C}$	[YPH21]
Ambient temperature	T_{amb}	22	$^{\circ}\text{C}$	lab estimate
Heat-loss coefficient	UA	18.0	W K^{-1}	eng. estimate
Proportional gain	K_p	50	W K^{-1}	eng. estimate
Heater saturation	Q_{max}	630	W	[Inf25]
Metabolic heat yield	$Y_{Q/X}$	12000	J g^{-1}	[CWM69]
min growth T	T_{min}	5	$^{\circ}\text{C}$	[Sal+11]
optimal T	T_{opt}	30	$^{\circ}\text{C}$	[Sal+11]
max growth T	T_{max}	40	$^{\circ}\text{C}$	[Sal+11]
Thermal time constant (derived)	τ_{th}	~ 5.2	min	$V\rho C_p/UA$
Steady-state T offset (derived)	ΔT_{ss}	~ 2.1	$^{\circ}\text{C}$	P-controller

the cardinal-pH model of Rosso et al. [Ros+95],

$$f_{\text{pH}}(\text{pH}) = \frac{(\text{pH} - \text{pH}_{\text{min}})(\text{pH} - \text{pH}_{\text{max}})}{(\text{pH} - \text{pH}_{\text{min}})(\text{pH} - \text{pH}_{\text{max}}) - (\text{pH} - \text{pH}_{\text{opt}})^2}, \quad (2.34)$$

which equals 1 at the optimum and 0 outside $(\text{pH}_{\text{min}}, \text{pH}_{\text{max}})$; it is re-anchored so that $f_{\text{pH}}(\text{pH}_{\text{set}}) = 1$ at the operating setpoint. The cardinal values used here are $\text{pH}_{\text{min}} = 2.5$, $\text{pH}_{\text{opt}} = 4.7$ and $\text{pH}_{\text{max}} = 8.0$. The pH of the broth itself is governed by three contributions and modeled by

$$\frac{d\text{pH}}{dt} = -\alpha_{\text{meta}} \mu X + \alpha_{\text{pump}} (F_B - F_A), \quad (2.35)$$

where $\mu = \mu_G + \mu_E$ is the specific growth rate and X the biomass concentration. The first term is the *metabolic acidification*: as the cells grow on glucose and ethanol they acidify the medium—chiefly through proton extrusion by the plasma-membrane H^+ -ATPase during sugar uptake and through the secretion of acidic metabolites and CO_2 [Kot+99]. The coefficient α_{meta} expresses how many pH units are lost per gram of biomass formed. The remaining two terms are the corrective base (F_B , raising pH) and acid (F_A , lowering pH) flows delivered by the titrant pumps, converted into a pH change by α_{pump} [SK86].

The pump flows are set by a proportional controller acting on the deviation from the setpoint,

$$u = K_c (\text{pH}_{\text{set}} - \text{pH}), \quad (2.36)$$

$$F_B = \min(u, F_{B,\text{max}}) \text{ if } u \geq 0, \quad \text{else } 0, \quad (2.37)$$

$$F_A = \min(-u, F_{A,\text{max}}) \text{ if } u < 0, \quad \text{else } 0, \quad (2.38)$$

so that a pH below setpoint calls the base pump and a pH above it calls the acid pump, each saturating at the maximum flow the physical pump can deliver. The conversion from titrant flow to pH change follows from the titrant concentration and the buffering capacity of the medium,

$$\alpha_{\text{pump}} = \frac{C_{\text{titrant}}}{1000 \beta_{\text{buf}} V}, \quad (2.39)$$

where β_{buf} is the buffer capacity (the resistance of the medium to pH change, low here because the medium used by [YPH21] is essentially water) and V the working volume. The parameter values are listed in Table 2.5.

Table 2.5: pH-layer model parameters (fault-detection plant).

Parameter	Symbol	Value	Unit	Source
Setpoint pH	pH_{set}	5.0	–	[YPH21])
Cardinal min/opt/max	–	2.5 / 4.7 / 8.0	–	[Ros+95]
Acidogenesis coeff.	α_{meta}	0.4	$\text{pH h}^{-1}/(\text{g L}^{-1} \text{h}^{-1})$	lab. estimate
Titration concentration	C_{titrant}	0.1	mol L^{-1}	H_2SO_4 titrant
Buffer capacity	β_{buf}	2.89×10^{-4}	–	water capacity
Pump conversion (derived)	α_{pump}	~ 0.256	pH mL^{-1}	Eq. 2.39
Proportional gain	K_c	120	$\text{mL h}^{-1}/\text{pH}$	eng. estimate
Pump saturation	$F_{A,\text{max}}, F_{B,\text{max}}$	210	mL h^{-1} (=3.5 mL/min)	[Inf25]

Dissolved oxygen.

Metabolic reactions proceed either aerobically or anaerobically; in aerobic metabolism oxygen acts as the terminal electron acceptor and is continuously consumed by the cells. *Saccharomyces cerevisiae* is a facultative anaerobe and can ferment sugar to ethanol without oxygen, but efficient *propagation*—the production of biomass—relies on aerobic respiration, which yields far more biomass per unit substrate than fermentation. An adequate supply of dissolved oxygen is therefore essential when the objective is to grow yeast, and its measurement and control have long been recognized as central to yeast propagation [SD61].

Oxygen, however, is only sparingly soluble in the culture broth, so it cannot simply be dissolved once at the start; it must be transferred continuously from the gas phase to the liquid as the cells consume it. In a stirred-tank bioreactor this transfer is achieved by two complementary mechanisms. *Aeration* is the sparging of air (or oxygen-enriched air) into the vessel, which introduces the gas as rising bubbles. *Agitation* is the mechanical stirring that breaks these bubbles into smaller ones and disperses them throughout the broth, thereby increasing the gas–liquid interfacial area and prolonging the bubble residence time, both of which accelerate oxygen transfer; this is the role of the impeller system described in Section 2.5.2.

The rate of oxygen transfer achieved by the combination of aeration and agitation is quantified by the volumetric mass-transfer coefficient $k_L a$, introduced in the following section [van79; Dor13].

The quantity the simulator must track is the amount of oxygen dissolved in the liquid phase of the vessel, propagated from its initial value. This dissolved oxygen is usually reported as the *dissolved-oxygen tension* (DOT), expressed as a percentage of saturation, where the saturation concentration C^* is itself fixed by the temperature and, through Henry's law, by the partial pressure of oxygen in the sparged gas. DOT is therefore not a fixed quantity: it evolves as the balance between the oxygen transferred into the liquid and the oxygen consumed by the cells. For the liquid-phase oxygen concentration C this balance reads

$$\frac{dC}{dt} = \underbrace{k_L a (C^* - C)}_{\text{OTR} - \text{supply}} - \underbrace{q_{O_2} X}_{\text{OUR} - \text{demand}}, \quad (2.40)$$

where C and C^* are the oxygen concentrations in the liquid phase and in equilibrium with the gas phase, respectively, and X is the (viable) biomass concentration [PR96].

The two terms carry opposite signs because they act in opposite directions. The *oxygen uptake rate*, $\text{OUR} = q_{O_2} X$, is the cellular demand and thus removes oxygen; q_{O_2} is the specific oxygen uptake rate [$\text{mg g}^{-1} \text{h}^{-1}$] — how fast a unit of biomass consumes oxygen — and X the biomass concentration [g L^{-1}], so OUR is expressed in [$\text{mg L}^{-1} \text{h}^{-1}$]. The *oxygen transfer rate*, $\text{OTR} = k_L a (C^* - C)$, is the supply: it is driven by the concentration gradient $C^* - C$ between the saturated interface and the bulk liquid — transfer is fastest when the broth is far from saturation and ceases as $C \rightarrow C^*$ — and scaled by the volumetric mass-transfer coefficient $k_L a$. The latter is itself the product of the liquid-film coefficient k_L [m s^{-1}], which sets how quickly oxygen crosses the gas-liquid interface, and the specific interfacial area a [m^{-1}], the total bubble surface per unit volume; their product has units of inverse time (here [h^{-1}]).

Physically, $k_L a$ encodes the effect of aeration and agitation: more aeration introduces more gas and hence more interfacial area, while stronger agitation shatters the bubbles into smaller ones, again increasing a . van 't Riet [van79] showed that the dominant correlating variables are the power input per unit volume P_g/V and the superficial gas velocity v_s , so that $k_L a$ follows a correlation of the form

$$k_L a = K \left(\frac{P_g}{V} \right)^a (v_s)^b. \quad (2.41)$$

- P_g/V — the *gassed* stirrer power per unit liquid volume, set by **agitation**. The ungassed power is $P_0 = n_i N_p \rho N^3 D^5$ for $n_i = 2$ six-blade Rushton turbines (additive power), power number $N_p = 5$, broth density $\rho = 1000 \text{ kg m}^{-3}$, stirrer speed N and impeller diameter D ; the gassed value is $P_g = 0.5 P_0$.
- $v_s = Q_{\text{air}}/A_{\text{cross}}$ — the superficial gas velocity, set by **aeration**: the air flow rate divided by the vessel cross-section $A_{\text{cross}} = \frac{\pi}{4} D_{\text{vessel}}^2$, with $D_{\text{vessel}} = 90 \text{ mm}$ (DN90).

- $K = 0.002$, $a = 0.7$, $b = 0.2$ — coefficients from [van79] for a *non-coalescing* (ionic) medium, appropriate for a salt-containing fermentation broth; the result is multiplied by 3600 to convert from $[\text{s}^{-1}]$ to $[\text{h}^{-1}]$.

Final model.

The final model is larger than the one described in Section 2.1.4: the EKF now carries eight states instead of five, $\mathbf{x} = [X, G, E, \mu_{\max,G}, \mu_{\max,E}, T, \text{pH}, \text{DO}]$, the three environmental variables being governed by the balances derived in the previous paragraphs. What remains to be specified is how these three states feed back onto growth — that is, how they modify the rate at which the cells consume substrate and produce or re-consume ethanol.

Temperature and pH act as *multiplicative rate modifiers*. Following the cardinal models of Rosso et al. [Ros+95], each is mapped to a dimensionless factor $f_T(T)$ and $f_{\text{pH}}(\text{pH})$ that equals one at the operating point and falls to zero at the edges of the growth range; their product

$$\varphi = f_T(T) f_{\text{pH}}(\text{pH}) \quad (2.42)$$

scales the specific growth rate. Hence, as either variable drifts from its optimum the growth slows, and once it leaves the cardinal interval $\varphi \rightarrow 0$ and fermentation is paused until the environment becomes growth-permissive again.

Dissolved oxygen enters differently, and deliberately so. Rather than scaling the overall rate, it governs the *fate* of the carbon: how much of the incoming glucose is respired and how much overflows into fermentation. Oxygen availability is expressed by a re-anchored Monod factor $f_{\text{DO}}(\text{DO}) = \text{DO}/(K_O + \text{DO})$, normalized to one at saturation [PR96]. The oxidative (respiratory) glucose uptake is capped by the cell's respiratory capacity and by the available oxygen,

$$q_{\text{ox}} = \min(q_G, C_{\text{RESP}} f_{\text{DO}}(\text{DO})), \quad q_{\text{fm}} = q_G - q_{\text{ox}}, \quad (2.43)$$

so that any glucose arriving faster than the oxygen-limited respiratory ceiling $C_{\text{RESP}} f_{\text{DO}}$ overflows into the fermentative branch q_{fm} [SK86]. The two branches build biomass with different yields (respiration being the more efficient), and the re-consumption of ethanol likewise requires oxygen; a fall in DO therefore shifts carbon from respiration towards fermentation. In short, the contribution of dissolved oxygen is precisely *how much glucose or ethanol can be oxidatively consumed at a given saturation*. Because $\text{DO} \gg K_O$ in healthy operation, $f_{\text{DO}} \approx 1$ and oxygen has no effect until a fault drives DO down towards K_O .

Fault Injection

This subsection develops the third step of the methodology (Section 2.5.2): fault injection and thresholding. *Fault injection* is the deliberate introduction of a fault into the ground-truth simulator in order to observe how the system's behavior changes. *Thresholding* is the subsequent analysis of these effects, aimed at drawing a line between the EKF innovation under healthy operation — where deviations arise only

from noise and the inherent uncertainty of the process — and the innovation under an actual fault, so that detection is (i) as fast as possible and (ii) subject to as few false alarms as possible, a false alarm being an alert raised while the system is in fact healthy.

The balance between these two objectives is a classical problem in change detection: lowering the threshold shortens the detection delay but raises the false-alarm rate, and vice versa [BN93]. The appropriate operating point depends on the relative cost of each error. In a domestic fire alarm, a missed detection is catastrophic while a false alarm is merely inconvenient, so the threshold is set very low — even the small amount of smoke from cooking triggers it. In the present case, where a fault may mean lost product (and, in medical or pharmaceutical settings, potentially lives), the threshold must be chosen accordingly; here it is placed at a deliberately balanced operating point between early detection and false alarms.

Fault set and implementation.

The fault scenarios investigated in this work, along with their corresponding implementation parameters, are detailed in Table 2.6. With the model in place, each fault is straightforward to implement by acting directly on the relevant actuator rather than adjusting the controller setpoint. For instance, for `heater_off` the heater power is forced to zero ($Q_{\text{heat}} = 0$), for `acid_stuck` the acid pump runs at full capacity irrespective of the current pH, and for `base_off` the base-pump gain is disabled ($K_c = 0$).

Table 2.6: Investigated Fault Cases and Implementation Parameters

Fault Category	Fault Case	Implementation Parameter / Value
Heater	heater_off	Power left = 0 ($Q_{\text{heat}} = 0$)
	heater_clog	Power left = 6%
Dosing	base_stuck	Power left = 210
	acid_stuck	Power left = 210
	base_off	Power left = 0 ($K_c = 0$)
	acid_off	Power left = 0
Agitation / Aeration	agitation_aeration_off	Power left = 0
	agitation_aeration_clog	Power left = 3% ($k_{La} = 0.08$)

Healthy reference.

The healthy system behaves as shown in Figure 2.13, and every fault signature below is read as a departure from this reference.

Thermal faults.

The result of the heater fault is shown in Figure 2.14: with the heater off, the temperature falls towards the ambient value of 22 °C. For the baseline strain this

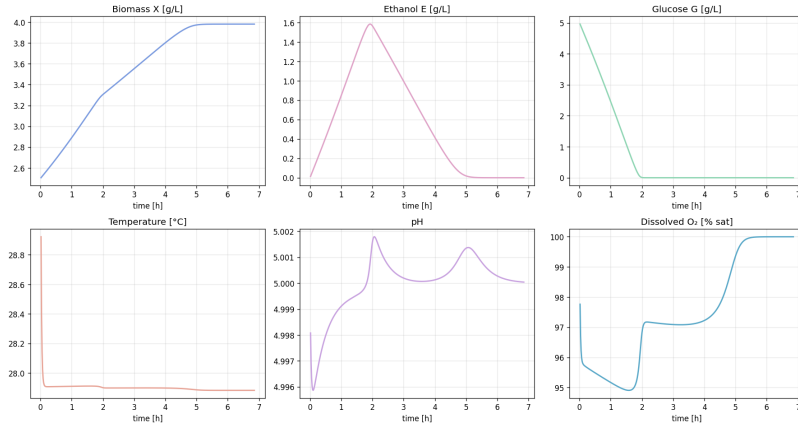


Figure 2.13: Healthy batch (reference behavior).

is below the optimum but still within the cardinal range, so growth continues, only more slowly. For a more temperature-sensitive strain, however, the same fault is far more damaging. Since this study aims at generality, the scenario was repeated for a hypothetical organism with a narrow cardinal range ($T_{\min} = 27$, $T_{\text{opt}} = 30$, $T_{\max} = 33$ °C); here 22 °C lies below $T_{\min}$, growth halts, and the final biomass is on average about 20 % lower than in the healthy (heated) case.

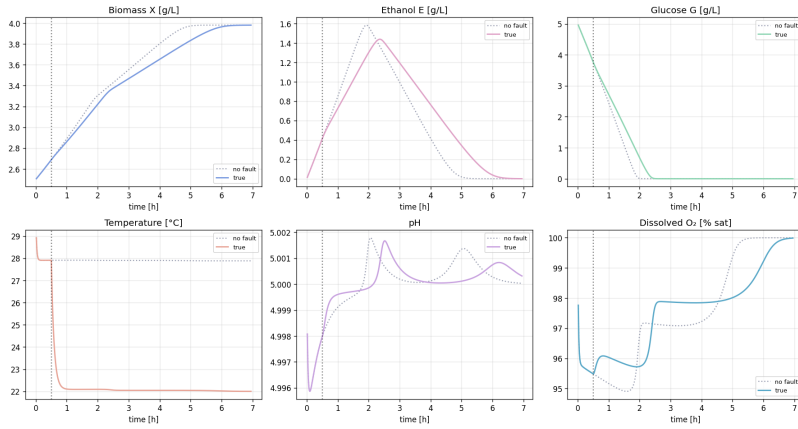


Figure 2.14: `heater_off` fault scenario. The faulted trajectory is shown against the healthy (fault-free) run, the black curves.

pH faults.

The result of `acid_stuck` is shown in Figure 2.15: the acid pump is stuck at full capacity. As long as the stuck flow stays below the opposing base pump's saturation flow, the base pump can pull the pH back towards the setpoint, settling at a sustained compensating flow — the detectable signature; beyond that, the pH crashes. The result of `base_off` is shown in Figure 2.16: because the only acidifying term left is the small metabolic acidification, the pH drifts only slightly and growth is barely

affected.

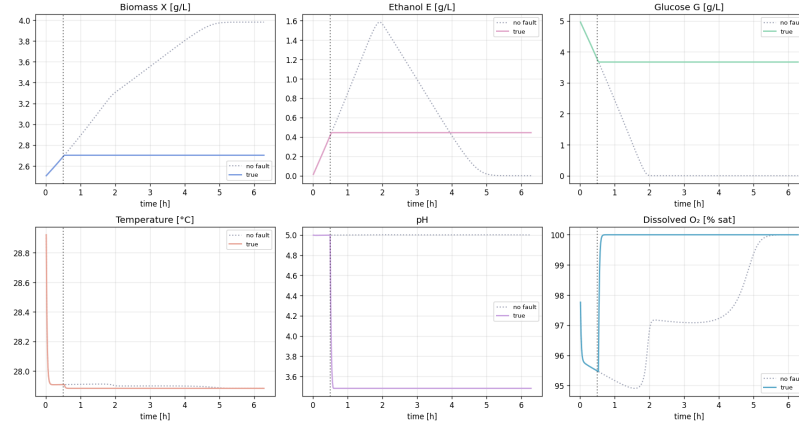


Figure 2.15: **acid_stuck** fault scenario. The faulted trajectory is shown against the healthy (fault-free) run, the black curves.

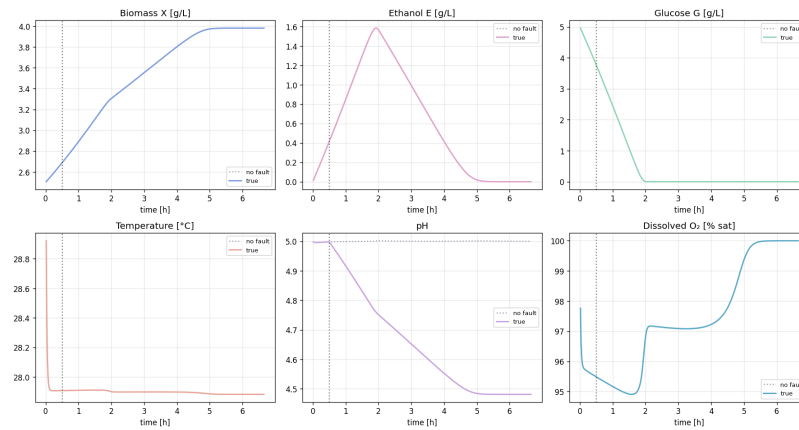


Figure 2.16: **base_off** fault scenario. The faulted trajectory is shown against the healthy (fault-free) run, the black curves.

Aeration/agitation faults.

Finally, the result of the agitation/aeration stop is shown in Figure 2.17. When stirring or aeration ceases, $k_L a \rightarrow 0$ and the dissolved oxygen is drawn down towards zero; respiration then shuts off and the carbon flux shifts entirely to fermentation. Growth does not stop outright — the cells continue to ferment glucose at a lower biomass yield — but the ethanol they produce can no longer be re-consumed, so residual ethanol accumulates and the final biomass is reduced. This rise in ethanol is the characteristic oxygen-fault signature discussed in the previous section.

Python Implementation.

The extended model is implemented in the Git Repository.

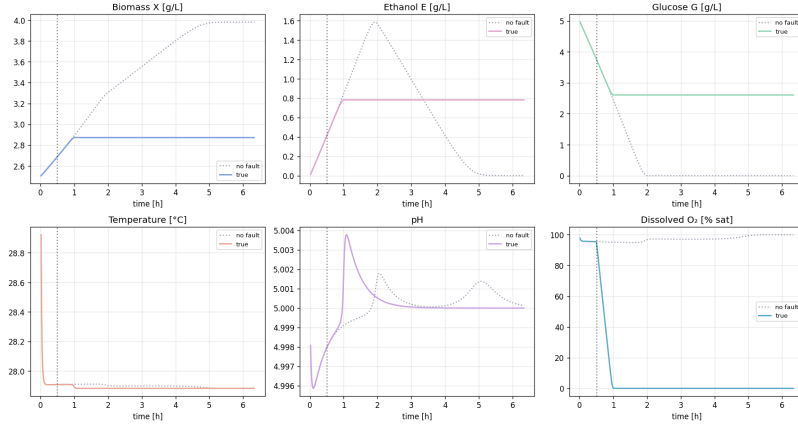


Figure 2.17: agitation_stop/aeration_stop fault scenario. The faulted trajectory is shown against the healthy (fault-free) run, the black curve.

Fault Identification

Sensor and actuator fault detection is an inherently difficult problem, because it must be addressed together with two confounding effects that are always present in a real estimator: the modeling errors between the actual system and its mathematical model, and the system and measurement noise [HC00]. Both push the innovation away from zero even when nothing is wrong, so the central difficulty is to separate a genuine fault from this ordinary, noise- and mismatch-driven variation.

Among the model-based FDI schemes, extended-Kalman-filter-based approaches are the most widely used, and a recurring pattern is to run not one but several filters in parallel: in a *multi-hypothesis* EKF, each filter embeds a particular fault hypothesis, and the bank as a whole detects and identifies the fault by comparing the filters' outputs [DSG14]. This bank-of-filters idea is the foundation of the isolation stage adopted here.

Building on these ideas, fault identification in this work is carried out in two stages. Detection is performed by monitoring the ethanol innovation of a single healthy-model EKF and testing it with a cumulative-sum (CUSUM) statistic, so that a sustained departure from the healthy reference triggers an alarm. RCA is then performed with a bank of extended Kalman filters — one per hypothesized fault — whose innovation sequences are compared in a Bayesian, multiple-model adaptive-estimation framework. The two stages are developed in the following sections. based on the research done in 2.5.1.

CUSUM.

In simple terms, the cumulative-sum (CUSUM) test raises an alarm when the recent measurements have become too surprising relative to the healthy model. At every five-minute step a new ethanol measurement z_k arrives; given the measurement-noise covariance R and the propagated state uncertainty P^- , the predicted ethanol $H\hat{x}^-$ is not expected to match z_k exactly, so a non-zero innovation $\nu_k = z_k - H\hat{x}^-$ is normal.

What matters is the *accumulation* of these innovations, normalized by the filter’s own predicted spread $\sqrt{S_k} = \sqrt{HP^-H^\top + R}$. Writing $\bar{\nu}_k = \nu_k/\sqrt{S_k}$, the two-sided CUSUM maintains

$$S_k^+ = \max(0, S_{k-1}^+ + \bar{\nu}_k - \kappa), \quad (2.44)$$

$$S_k^- = \max(0, S_{k-1}^- - \bar{\nu}_k - \kappa), \quad (2.45)$$

and fires when either arm exceeds the decision interval h . The S^+ arm accumulates evidence that ethanol runs above the prediction and S^- that it runs below (the fault direction). The reference value (slack) κ subtracts a small amount at each step, so the ordinary random walk of the noise does not accumulate across h on its own; the $\max(0, \cdot)$ likewise stops the arms from drifting to spurious negative values.

Calibrating κ and h .

The two parameters were fixed against a healthy ensemble of **30** independent noise realizations (30 seeds $\times$ 84 samples). Over this ensemble the innovation has a mean of +0.0016 g/L (essentially zero), an empirical standard deviation of 0.051 g/L, and a lag-1 autocorrelation of -0.04 , i.e. it is essentially white, so an i.i.d. CUSUM design is valid. The slack was set to $\kappa = 0.15$, half of the normalized interruption level $\bar{\nu}_{0.1} = (0.1 \text{ g/L})/\sqrt{S} \approx 0.30$ — that is, half of the 0.1 g/L ethanol deviation deemed worth reacting to. With κ fixed, the CUSUM was run over the same 30 healthy realizations; the largest arm reached on any healthy batch was approximately 0.5. The decision interval was therefore set to $h = 1.5$, comfortably above this healthy maximum so that no healthy batch raises an alarm, in line with the detectability considerations of Section 2.5.4 [Pag54; BN93].

It should be noted that the measurement-noise covariance is inherited from [YPH21] ($R = 0.1 \text{ g}^2/\text{L}^2$), whereas the true sensor noise is far smaller ($\approx 0.0025 \text{ g}^2/\text{L}^2$); consequently $\sqrt{S}$ overstates the innovation spread by roughly a factor of six, and one real standard deviation corresponds to only 0.15 in the units the CUSUM sees. In these units $\kappa = 0.15$ is effectively a one-sigma slack, which makes the detector deliberately conservative — the healthy arms remain at zero and only faults that clearly arrest growth trigger an alarm.

Python Implementation.

The detection and isolation stages were implemented in Python, using NumPy [Har+20] for the array algebra, SciPy [Vir+20] for the numerical routines, and Matplotlib [Hun07] for the figures. Both stages reuse the single EKF implementation of the digital shadow, so every result below is attributable to the model rather than to a divergent re-implementation.

The **CUSUM** detector is deliberately lightweight: it carries only two scalar accumulators, S^+ and S^- , updated once per correction step from that step’s normalised innovation (Eqs. above), and raises an alarm as soon as either exceeds the decision interval h . The **MMAE** isolator is a bank of extended Kalman filters in which each

candidate fault is injected into the filter’s own process model — a context manager temporarily rebinds the model so that the filter “believes” its hypothesised fault, without creating a second simulator — and the filters are scored by the Bayesian recursion of Eq. (2.48), accumulated from the estimated onset.

The CUSUM method is implemented in the Git Repository.

Root Cause Analysis.

While the traditional RCA methods — brainstorming, the five-whys, the Ishikawa diagram — introduced in Section 2.5.1 remain popular and widely used in practice for their structured, easy-to-use character, they can be insufficient for complex, data-rich problems [MLT21]. In the present setting every fault ultimately have the same effect on the growth, it either stops or slows down, so the causes are not trivially separable. Which means several faults leave almost identical ethanol traces. Among the model-based alternatives, multiple-model adaptive estimation (MMAE) has been used successfully in many comparable problems in combination with an EKF [HM00; VCM12]; it is therefore the natural choice here (Section 2.5.1).

The EKF uses the measurement-noise covariance of Yousefi-Darani, Paquet-Durand, and Hitzmann [YPH21], $R = 0.1$, which is relatively large: it makes the filter trust its model — the healthy expectation — more than the individual measurement. This is deliberate. If the filter trusted the measurement too much, it would, under a fault, simply follow the faulty reading and adapt its own prediction to it, so that the fault would vanish from the innovation; keeping R large forces the fault to appear in the innovation instead. The behavior of the filter under a failure, here agitation/aeration stopped working, is shown in Figure 2.18.

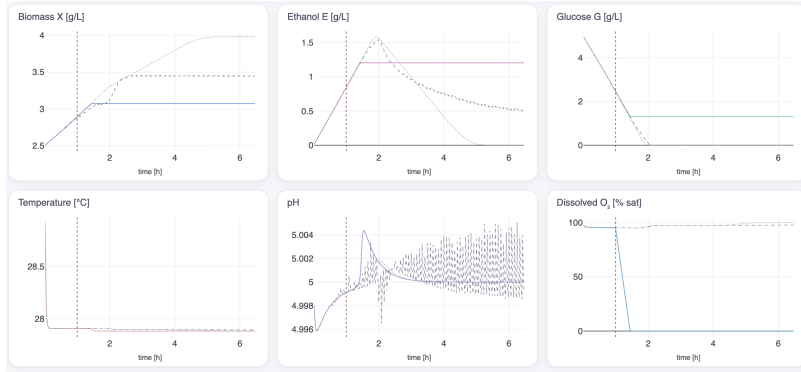


Figure 2.18: EKF behavior in the `agitation_stop/aeration_stop` fault scenario. The solid curve is the healthy (fault-free) trajectory; the dotted curve is the EKF estimate.

After a fault has been detected — the CUSUM arm crossing h raises the alarm, and the arm is walked back to estimate the onset time — a bank of extended Kalman filters is run in parallel, each embedding one candidate fault from the catalogue, started at the estimated onset. The central question is: given the real measurements, which hypothesis is most probable? This is answered by the *likelihood*. Each filter

j produces, at every step, a Gaussian prediction of the ethanol concentration with mean $\hat{z}_j = H\hat{x}_j^-$ (the ethanol component of its state estimate) and variance S_j — the filter’s uncertainty about its own prediction, which grows between measurements and shrinks at each update. When a new measurement z arrives, its height under that Gaussian is the likelihood of the measurement under hypothesis j (Figure 2.19); the further the measurement is from the predicted peak $\hat{z}_j$, the lower it sits on the curve. Writing the innovation $\nu_j = z - \hat{z}_j$, this likelihood is

$$\mathcal{L}_j = \mathcal{N}(z; \hat{z}_j, S_j) = \mathcal{N}(\nu_j; 0, S_j). \quad (2.46)$$

A filter whose assumed fault matches reality produces small innovations and hence a high likelihood, whereas a mismatched filter produces large innovations and a low likelihood — this is the backbone of the isolation. By Bayes’ rule the posterior probability of hypothesis j given the measurements is proportional to its likelihood times a prior,

$$p(H_j | z_{1:k}) \propto p(z_{1:k} | H_j) p(H_j). \quad (2.47)$$

Because there is no reason to favour any fault before seeing data, a uniform prior is assigned, $p(H_j) = 1/8$ over the eight hypotheses in the bank; the constant prior then cancels in the normalisation, so the posterior reduces to the normalised likelihood. Applied recursively over the samples from the estimated onset this gives

$$p_j(k) \propto p_j(k-1) \cdot \mathcal{N}(\nu_j(k); 0, S_j(k)), \quad (2.48)$$

and the hypothesis with the highest posterior identifies the root cause. In practice the recursion is evaluated in log-space to avoid numerical underflow when many small Gaussian likelihoods are multiplied: the per-sample log-likelihoods are summed from the onset and the posterior is recovered by a *softmax* (a normalised exponential) over the accumulated log-likelihoods. With the uniform prior this is identical to the Bayesian update above.

The RCA method is implemented in the Git Repository.

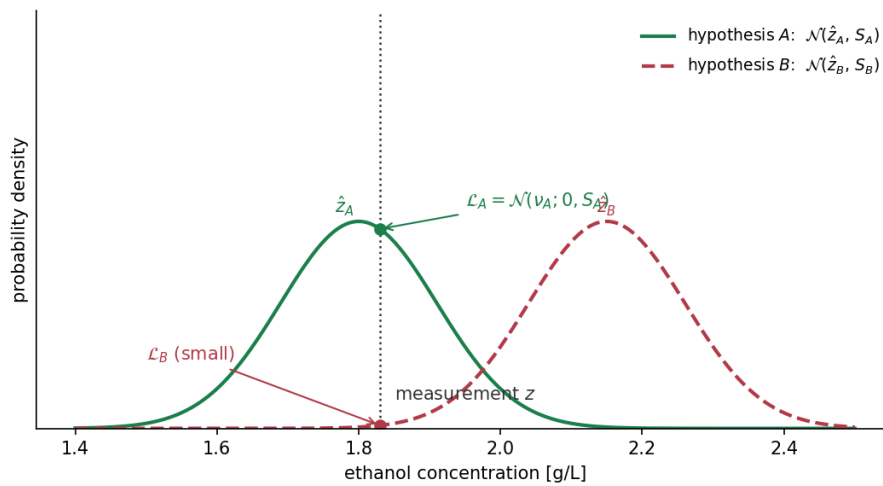


Figure 2.19: MMAE likelihood. Each hypothesis filter predicts a Gaussian over ethanol, $\mathcal{N}(\hat{z}_j, S_j)$. The same measurement z sits high on the curve of the filter whose assumed fault matches reality (large likelihood $\mathcal{L}_A$) and low on a mismatched one ($\mathcal{L}_B$); the posterior of Eq. (2.48) accumulates these into the diagnosis.(generated by [Ant25])

3 Results and Analysis

Detection, onset and delay.

Detection and isolation are separated by an intermediate step that recovers *when* the fault began. When the alarm fires at sample i , the arm that crossed the threshold is selected (S^- if $S_i^- \geq S_i^+$, otherwise S^+) and walked backwards to the last sample at which it was still zero; that index is taken as the estimated onset $\hat{t}_{\text{fault}}$, from which the MMAE bank is then scored. For the heater fault used as the running example, the true onset is at 0.5 h; the alarm fires at 1.0 h, i.e. a detection delay of about 30 min, and the backtracked onset estimate is 0.42 h, only about 5 min early. The onset estimate is deliberately approximate — for slow faults whose early innovations fall below the CUSUM slack it can lag by considerably more (up to about 65 min for the slowest cases) — but it is accurate enough to anchor the Bayesian scoring window. On this example the MMAE bank then assigns a posterior probability of 0.234 to the correct `heater_off` hypothesis.

Detected Failure example (user interface).

Figures 3.1–3.3 show the three stages of this heater-fault example as presented in the user interface: the CUSUM detection and estimated onset, the parallel hypothesis bank, and the resulting posterior that identifies the fault.

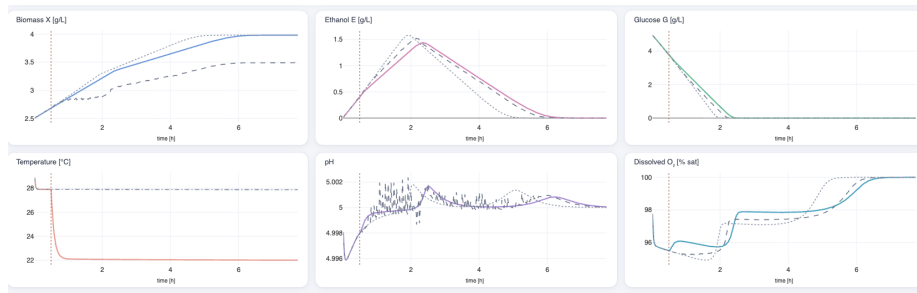


Figure 3.1: Heater fault — scenario: the heater stopped working after 30 minutes, the temperature will fall to the laboratory’s temperature and the cultivation slows down. The solid curve is the healthy behavior; the dotted curve is the EKF

Figures 3.4–3.6 show the three stages of this aeration/agitation-fault example as presented in the user interface: the CUSUM detection and estimated onset, the parallel hypothesis bank, and the resulting posterior that identifies the fault.



Figure 3.2: Heater fault — detection: the ethanol innovation and the CUSUM arms; the arm crosses the decision interval h (alarm), and is walked back to the estimated onset.



Figure 3.3: Heater fault — the posterior probability over the hypotheses, which identifies `heater_off` ($p \approx 0.234$).

Figures 3.7–3.9 show the three stages of this acid-pump-stuck-fault example as presented in the user interface: the CUSUM detection and estimated onset, the parallel hypothesis bank, and the resulting posterior that identifies the fault.

Python Implementation

Other fault cases as well as the results reported in this section are available and reproducible using Git Repository.

Additional Sensors.

Ethanol is not the only measurable state: temperature, pH and dissolved oxygen can each be instrumented and added to the filter as an extra measurement row in H , carrying its own innovation and detection threshold. Whether a sensor is worth its cost splits into two distinct questions. The first is an *estimation* question — is the process model accurate enough for the EKF to reconstruct the state without the sensor? — which requires the *validation* that was performed by [YPH21]: running the EKF on the process model while collecting off-line reference data and comparing the two. Because the present work uses purely simulated data, that comparison could not be carried out here. The second is a *detection* question — does the sensor expose faults the ethanol channel cannot? — which the simulation can address directly.

The multi-channel study 4.2 adds temperature, pH and dissolved-oxygen signals as further measurement rows and compares five measurement systems — ethanol

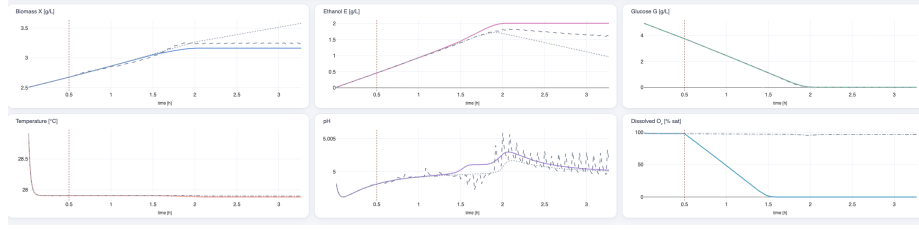


Figure 3.4: Aeration/Agitation fault — scenario: the aeration/agitation system stopped working after 30 minutes, the DO will fall and cultivation continues until glucose is depleted, after which growth ceases because yeast cannot metabolize ethanol under anaerobic conditions 2.5.3. The solid curve is the healthy behavior; the dotted curve is the EKF



Figure 3.5: Aeration/Agitation fault — detection: the ethanol innovation and the CUSUM arms; the arm crosses the decision interval h (alarm), and is walked back to the estimated onset.

alone (E), ethanol plus one probe (E+T, E+pH, E+DO) and the full rack (E+all) — with every channel sampled at the same five-minute cadence as ethanol and probe noise $\sigma_T = 0.1^\circ\text{C}$, $\sigma_{\text{pH}} = 0.02$ and $\sigma_{\text{DO}} = 1\%$ saturation. Detection is a bank of independent two-sided CUSUMs, one per channel, the first arm to cross raising the alarm; the ethanol arm keeps the deployed pair $(\kappa, h) = (0.15, 1.5)$ and each probe arm the σ -equivalent $(1.0, 10.0)$, so that no threshold is re-tuned per instrument and the comparison measures the sensor rather than its calibration.

The result is that a sensor only helps the faults that move its own variable, and the split follows the control loops.

3 Results and Analysis



Figure 3.6: Aeration/Agitation fault — the posterior probability over the hypotheses, which identifies `Aeration/Agitation_stop` ($p \approx 0.709$).

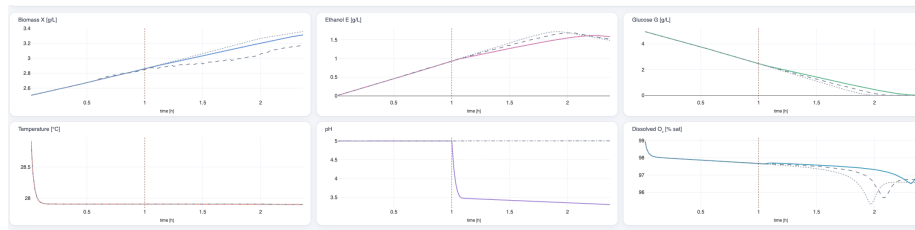


Figure 3.7: Acid-pump-stuck fault — scenario: the acid pump stuck after 1 hour, the pH will fall and the cultivation slows down. The solid curve is the healthy behavior; the dotted curve is the EKF



Figure 3.8: Acid-pump-stuck fault — detection: the ethanol innovation and the CUSUM arms; the arm crosses the decision interval h (alarm), and is walked back to the estimated onset.



Figure 3.9: Acid-pump-stuck fault — the posterior probability over the hypotheses, which identifies `Acid-pump-stuck_on` ($p \approx 0.257$).

4 Discussion

Evaluation

The results reported above concern only faults that affect yeast growth — cases in which the culture consumes or produces substrate more slowly than the healthy model predicts, so that the mismatch surfaces in the ethanol innovation. A fair evaluation must make the method’s limitations explicit, since detectability is a property of the plant, not of the estimator.

Detectability depends on how sensitive the assumed strain is to its environment. The reported runs were also carried out on the *sensitive* strain, for which growth ceases below $T_{\min} = 27^{\circ}\text{C}$, whereas under the *robust* strain $T_{\min} = 5^{\circ}\text{C}$. A thermal fault such as the heater running at half power, which leaves growth untouched on the robust strain, can therefore halt it entirely on the sensitive one. On ethanol alone the detector is blind to any fault that does not propagate into growth — in the current registry `acid_off` and `base_off`. This is partly acceptable, since the aim is to catch faults that interrupt the cultivation, but it is also a genuine limitation: a fault that is silent under one strain, medium or operating point may degrade the process under another, and it is then that the value of a dedicated sensor becomes apparent.

The system developed in this work is an EKF-based digital shadow able to detect process anomalies with an average detection delay of 72 min, followed by cause identification that takes on average 172 min. Adding the reactor’s own temperature, pH and dissolved-oxygen probes reduces the detection delay to 11 min and settles the cause within 20 min. The overall evaluation is summarised in Table 4.1: the detector and the MMAE isolator were run on all eight faults of the catalog on a single noise realization, first on the ethanol channel alone and then with the probes added to the likelihood.

How many faults are detected, and is their verdict correct?

On ethanol alone, six of the eight faults are detected, and every one of those six is then identified correctly. The two that are missed are `acid_off` and `base_off` with less effect on the cultivation.

How long does it take?

Detection is not the bottleneck; isolation is. The mean delay from onset to the alarm is about 72 min, but a correct, stable verdict takes on average close to three hours, dominated by the slow pump faults — `base_stuck` alone needs 335 min. The alarm

Table 4.1: Detection and diagnosis over the eight-fault catalog. Each cell gives the ethanol-only value and the value with the T/pH/DO probes added (eth $\rightarrow$ probes). “—” = not detected. Times are minutes after fault onset; the totals are means over the detected scenarios.

Fault	detected?		alarm [min]	correct verdict [min]
	ethanol	+probes	eth $\rightarrow$ probes	eth $\rightarrow$ probes
Heater off	yes	yes	40 $\rightarrow$ 5	160 $\rightarrow$ 5
Heater clog	yes	yes	45 $\rightarrow$ 5	45 $\rightarrow$ 35
Acid pump off	no	no	—	—
Acid pump stuck	yes	yes	40 $\rightarrow$ 5	160 $\rightarrow$ 5
Base pump off	no	yes	— $\rightarrow$ 40	— $\rightarrow$ 40
Base pump stuck	yes	yes	35 $\rightarrow$ 5	335 $\rightarrow$ 5
Agitation/aeration off	yes	yes	85 $\rightarrow$ 10	85 $\rightarrow$ 10
Agitation/aeration clog	yes	yes	185 $\rightarrow$ 10	245 $\rightarrow$ 40
totals	6/8	7/8	72 $\rightarrow$ 11	172 $\rightarrow$ 20

fires as soon as the ethanol innovation becomes improbable, which is well before the deviation has a recognizable shape.

What do the probes change?

The probes help in two distinct ways. First, they *detect* a fault ethanol never sees: **base_off** leaves no ethanol signature but shifts the pH directly, so detection rises from six to seven of eight. Second, they collapse the timing — the mean alarm delay falls from 72 to 11 min and the mean settled verdict from 172 to 20 min, because the instruments report which quantity moved and in which direction, a single sample’s worth of evidence, whereas ethanol must accumulate a change in the shape of the trajectory. The pH probe recovers **base_off** and it also collapses the alarm on the remaining pH-loop faults, from 40 to 5 min on **acid_stuck** and from 35 to 5 min on **base_stuck**. The temperature probe does the same for the thermal loop, cutting the two heater alarms from 40 and 45 min to 5 min, and the DO probe for the aeration loop, cutting the two agitation alarms from 85 and 185 min to 10 min. Outside its own loop each probe is silent: pH gives no earlier warning for the heater or the agitation faults, temperature none for the pump faults, and DO none for the heater or pump faults — in every one of those cases the alarm is still raised by the ethanol arm, at the ethanol-only delay. A probe outside the fault’s loop can even cost a little, since it contributes noise but no evidence to the likelihood: adding DO alone delays the settled verdict on **heater_clog** from 45 to 75 min.

It should be stressed that the probes are monitors and are never fed back into the shadow: feeding a measured temperature, pH or dissolved oxygen into the filter would let it follow the fault and go blind on it, and every gain reported here depends on the shadow continuing to integrate healthy dynamics.

Why `base_off` is silent on ethanol.

The only term that lowers the pH is the metabolic acidification, which is too weak to drive the pH below its growth-limiting minimum quickly, so the cultivation proceeds essentially as a healthy one and nothing reaches the ethanol channel. This is a limit of detectability from the ethanol measurement, not a shortcoming of the estimator — and it is exactly the gap the pH probe closes.

Which fault no probe can help.

`acid_off` is undetectable on every channel, and this is a property of the plant, not of the sensing: yeast metabolism only ever drives the pH downward, so the controller never calls for acid, and a dead acid pump is asked to do nothing — the batch is identical to a healthy one in every state. No instrument can reveal a fault that leaves no trace.

The implemented user interface (Section 2.4.1) lets the user change all initial conditions as well as the cardinal (minimum and maximum) temperatures and pH, so that the same scenarios can be re-examined for more sensitive strains. The interface is also designed to host more than one bioreactor — at present only the Minifors 2 is implemented, parameterized from its manual [Inf25], but further reactors can be added and selected — and, likewise, more than one organism: only the baker’s-yeast cultivation is implemented so far, but the backend is structured so that, with sufficient domain knowledge, other cells or reactions can be added. Combined with biological expertise, such a tool has the potential to make a substantial impact wherever a faithful, failure-free representation of a living system is required.

Fair Evaluation.

A comprehensive evaluation of the proposed method will require validating the process model introduced in Section 2.5.3 against domain expertise and experimental data from a physical laboratory setup. Addressing the current reliance on simulated data represents both the primary limitation of this thesis and the key direction for future research.

Python Implementation

The tables and the reported values are also available in the Git Repository.

A General Road-Map

One of the central contribution of this work is a reusable road-map for building a digital shadow of a simple bioreactor with fault-detection and root-cause-analysis capabilities. Its steps are:

1. **Define the states.** Identify the quantities to be estimated — here biomass, glucose, ethanol, temperature, pH and dissolved oxygen, together with the maximum growth rates on glucose and on ethanol (eight states in total).

2. **Model their dynamics.** Specify the initial value of each state and a differential equation describing how it evolves over time.
3. **Set the growth limits.** Fix the cardinal minimum, optimum and maximum temperature and pH for the organism; where organism-specific values are unavailable, adopt the cardinal-model form of Rosso et al. [Ros+95].
4. **Gather the reactor data.** Establish the hardware parameters: heater power, wall construction, the acid and base used and their flow rates, the number and speed of the impellers, and the aeration rate.
5. **Decide the instrumentation.** Determine what can be measured and how often, and how additional sensors would improve the estimate — as in the cadence experiment of Table 2.1 — trading sensor cost against simplicity. If possible this decision can be best made using the same validation process used by [YPH21] which is comparing the real sensors value to the estimated values, generated by the digital shadow only using the process model.
6. **Enumerate the faults.** Identify the plausible fault cases.
7. **Inject and screen.** Using the simulator, inject each fault and test whether it is separable by simple means, establishing the boundary between healthy and faulty behavior.
8. **Tune the detector.** Calibrate the CUSUM slack and decision interval based on the current use case.
9. **Isolate the cause.** Apply the MMAE method using a bank of parallel filters to attribute the detected fault to its cause.

5 Conclusion

Summary of the Thesis

This work provides a roadmap for creating a digital shadow of a simple bioreactor, with each step investigated through a concrete implementation using baker’s yeast (*Saccharomyces cerevisiae*). Building upon the model of [YPH21], this thesis expands the framework in three primary directions: first, by evaluating the model’s limitations (2.3.3-2.4); second, by extending the 5-state Extended Kalman Filter (EKF) to an 8-state formulation (2.5.3); and finally, by integrating a fault detection and identification mechanism on top of the estimator (2.5.5). These contributions were realized by synthesizing methodologies from existing literature and developing novel solutions to address technical challenges encountered throughout each phase.

Challenges and Limitations

The two principal limitations of this work are the lack of biological domain expertise — which constrains the completeness of the fault catalog — and the absence of real cultivation data, so that the system is validated against models rather than physical reality. In addition, several parameters are engineering estimates rather than measured values, since there was no opportunity to measure them in a laboratory.

Future Work

Several extensions follow naturally. The framework can be broadened by adding further bioreactors, organisms and reactions and by assembling the corresponding data into a reusable database. The parameters currently estimated could be replaced by expert-provided or experimentally measured values, and the whole system validated on real cultivation data. Another direction would be expert validation of the mathematical models introduced for pH, temperature, and dissolved oxygen (DO). Future work should also involve a comprehensive investigation of failure modes, starting with expert validation and completion of the model’s parameters followed by experimental testing. Finally, where the fault cases become harder to separate, the model-based MMAE isolation could be replaced by data-driven, machine-learning classifiers — such as random forests or decision trees — trained on the (augmented) simulation data [MLT21].

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Statutory Declaration

I hereby certify that I have written this thesis independently and without outside help. I have not used any aids or sources other than those specified by me and I have marked the passages taken from the works used in terms of content and wording as such. I confirm that the electronic version submitted is identical to the printed copies.

In preparing this thesis I used generative AI tools (Claude Opus 4.8 and Claude Code, Anthropic) for the following purposes: language editing of English text passages, assistance with the development and debugging of the Python implementation, and support in searching for literature.

The conceptual design, the modelling, the design, execution and evaluation of the experiments, and the scientific conclusions are my own work. All output produced by AI tools was reviewed for correctness; all bibliographic references were verified against the original sources for existence and accuracy. I take full responsibility for the entire content of this thesis.

Rostock, 07. 09. 2026



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